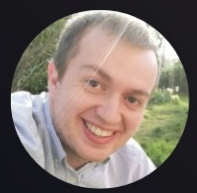
Simulation-based inference for exoplanets



Gilles Louppe
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Max Dax
maximilian.dax@tuebingen.mpg.de

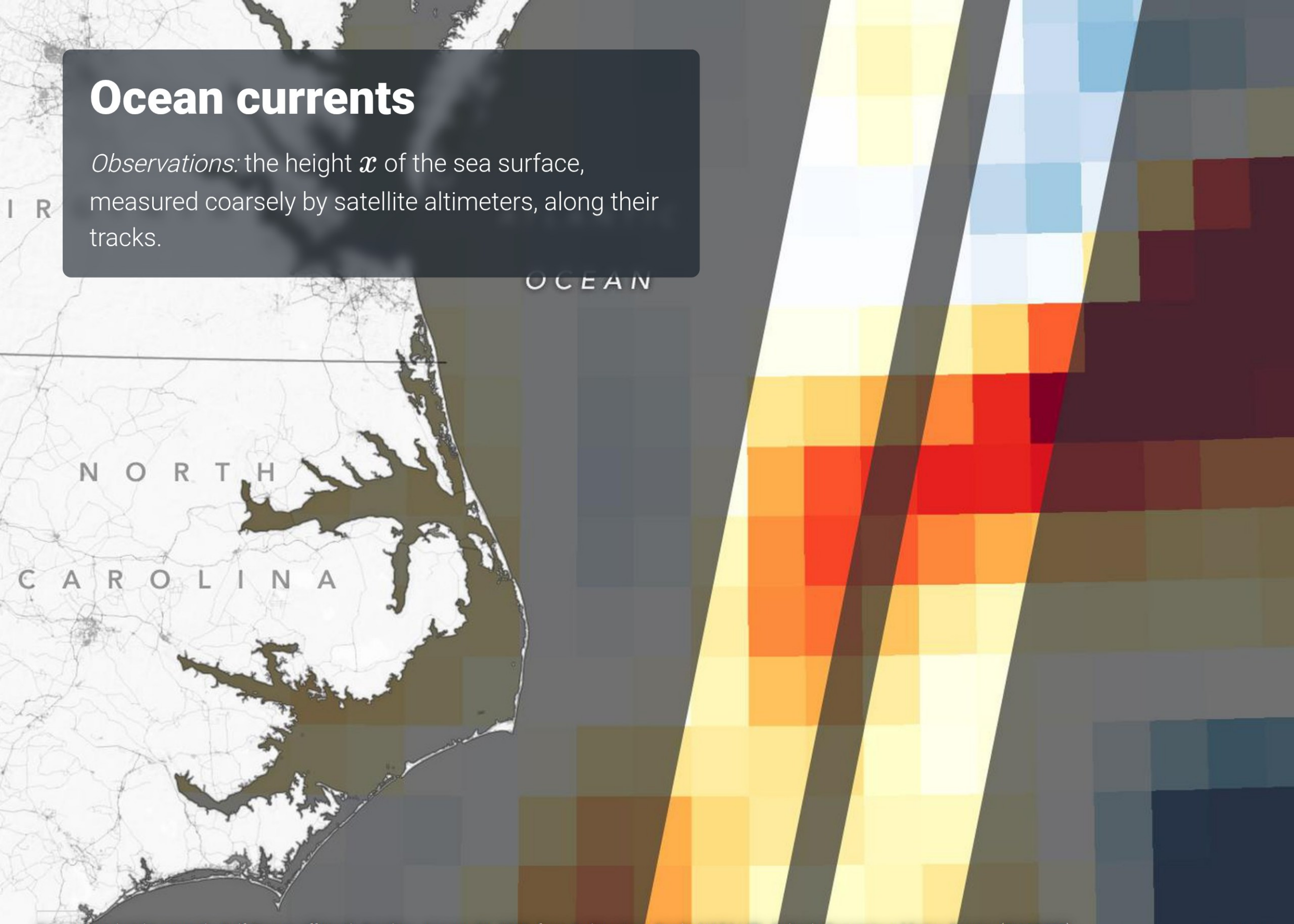
Exomints 2026
November 10, 2026

ellis | INSTITUTE TÜBINGEN LIÈGE université

Illustration of WASP-39 b: NASA, ESA, CSA, J. Olmsted (STScI).

Ocean currents

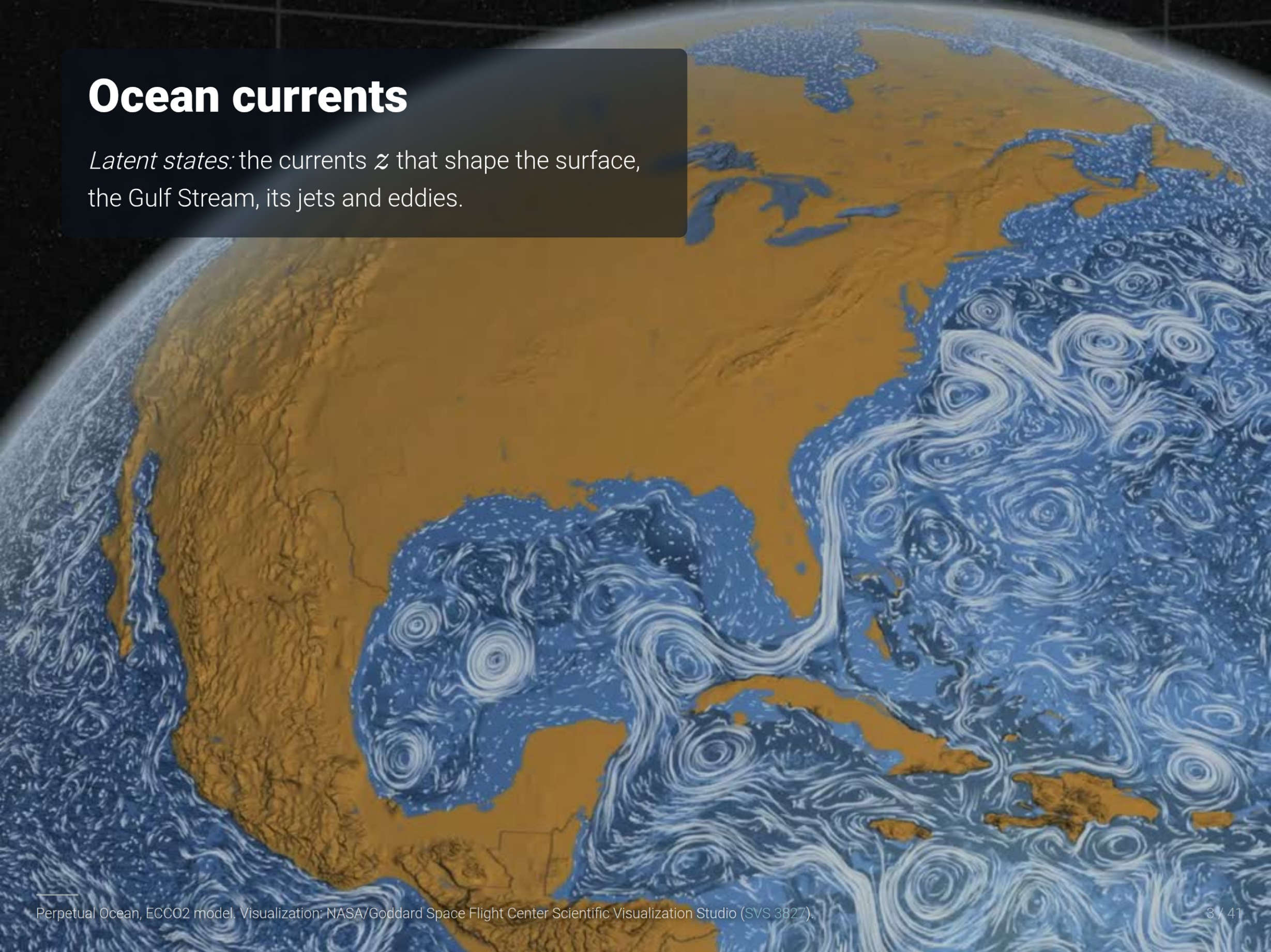
Observations: the height x of the sea surface, measured coarsely by satellite altimeters, along their tracks.



Sea surface height anomaly, Gulf Stream off North Carolina, January 21, 2023, from 7 altimeters. Credit: NASA/JPL-Caltech/Copernicus Marine Service (PIA25773).

Ocean currents

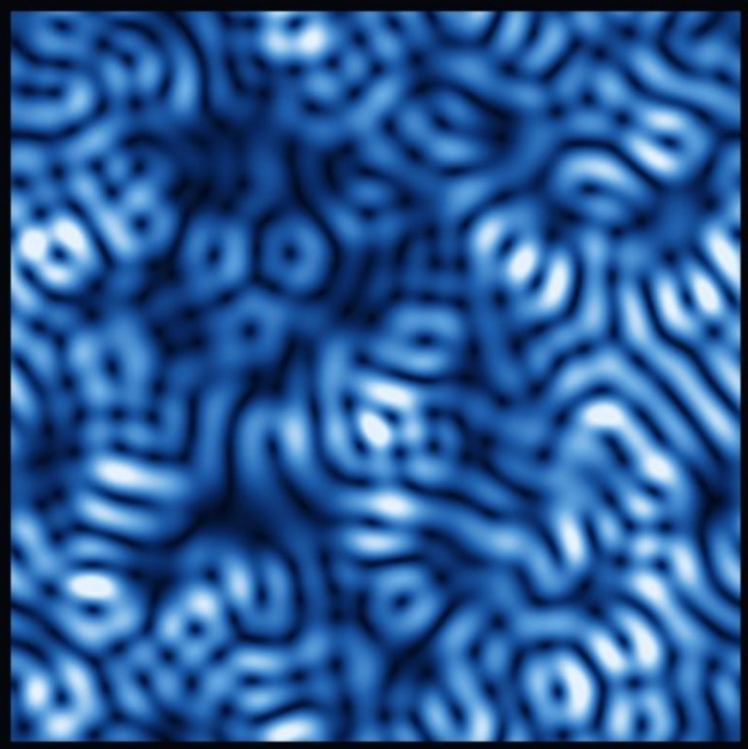
Latent states: the currents z that shape the surface, the Gulf Stream, its jets and eddies.



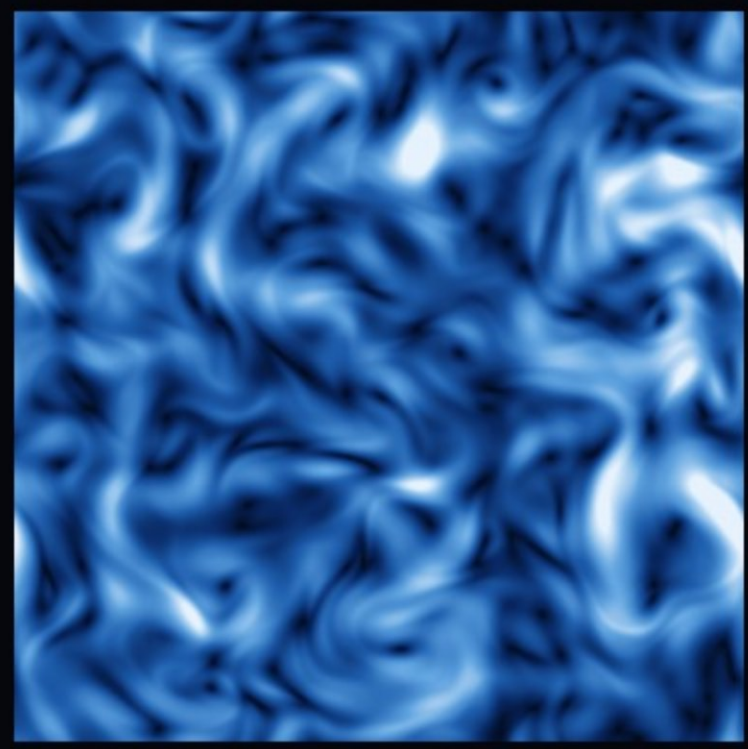
Perpetual Ocean, ECCO2 model. Visualization: NASA/Goddard Space Flight Center Scientific Visualization Studio (SVS 3827).

Ocean currents

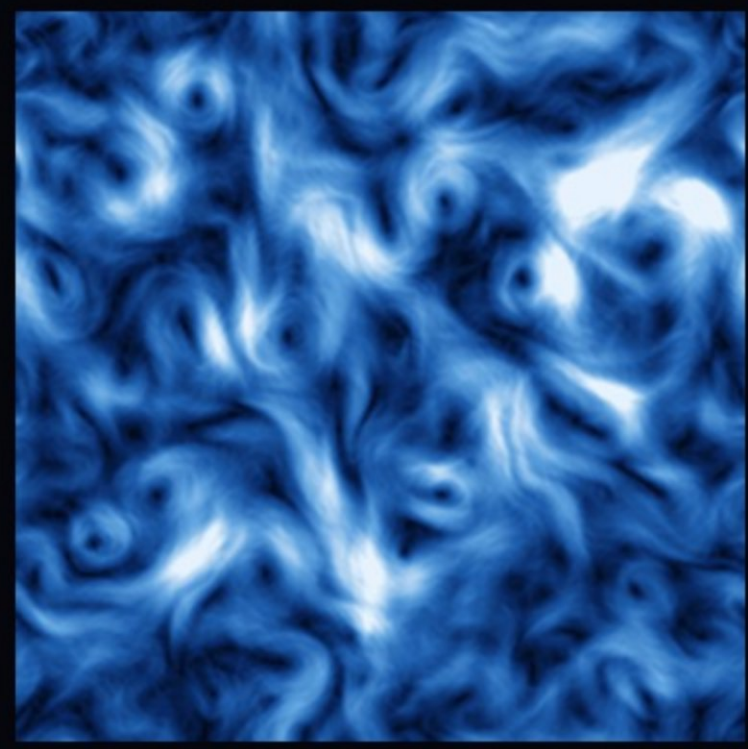
Parameters: the global constants θ of the ocean model, such as its viscosity ν. Same equations, different values, different oceans.



$\nu = 10^{-2}$



$\nu = 10^{-3}$



$\nu = 10^{-4}$

Illustration: 2D turbulence on a β-plane, same forcing, three viscosities; current speed.

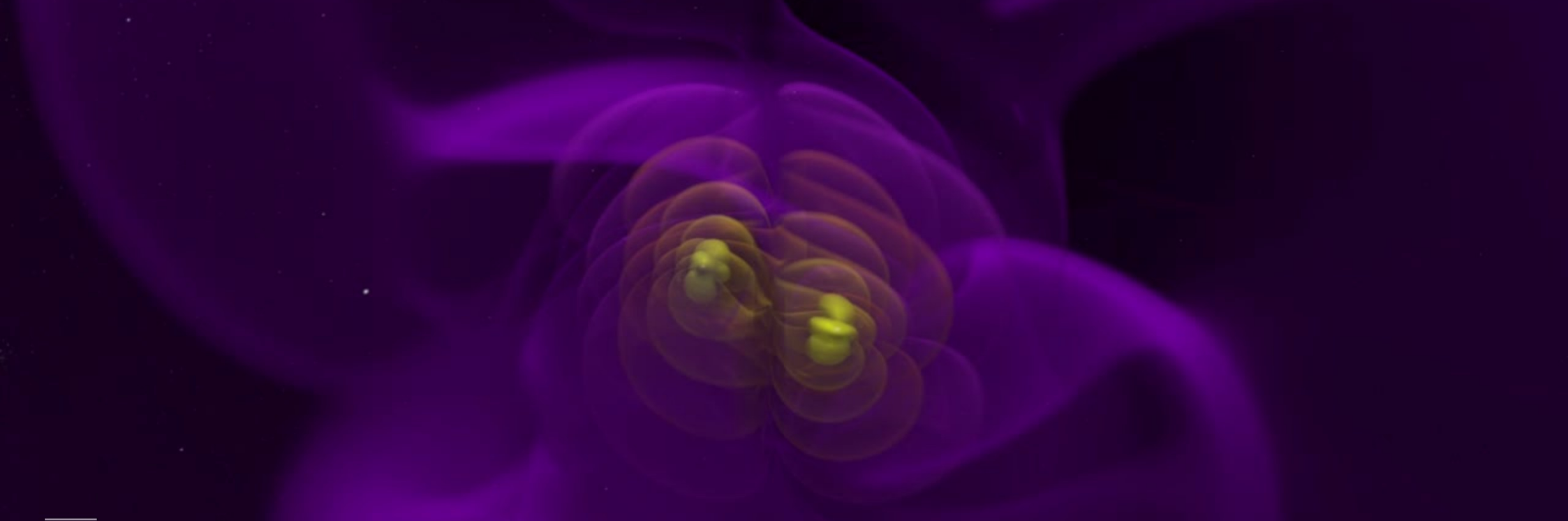
Gravitational waves

Observations: the strain x recorded by LIGO. In time and frequency, a chirp rises out of the noise.

Data: LIGO Hanford, GW150914, September 14, 2015 (GWOSC). Whitened strain, band-passed 35–350 Hz (white), and its wavelet transform, 20–512 Hz.

Gravitational waves

Latent states: the signal z of two black holes that spiral into each other, merge, and ripple space-time.



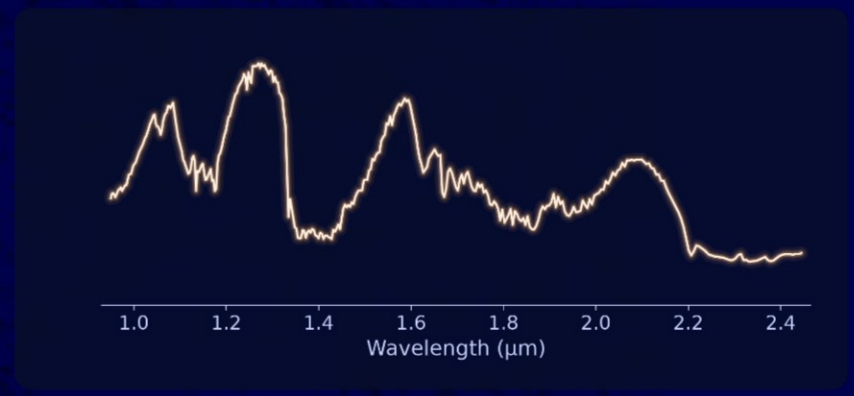
Simulation: B. J. Kelly, C. Henze, T. Sandstrom, NASA SVS 13197.

Gravitational waves

Parameters: the properties θ of the binary, masses, spins, distance, orientation, sky position (15 in total).



Illustration: LIGO/Caltech/MIT/Sonoma State (Aurore Simonnet).



Exoplanets

Observations: the spectrum $\boldsymbol{x}$ of a planet, whose light direct imaging separates from its star.

HR 8799, Keck, 2009–2016: Jason Wang (Caltech), Christian Marois (NRC Herzberg), CC BY 3.0. Spectrum: simulated benchmark of Vasist et al, A&A 2023.

Exoplanets

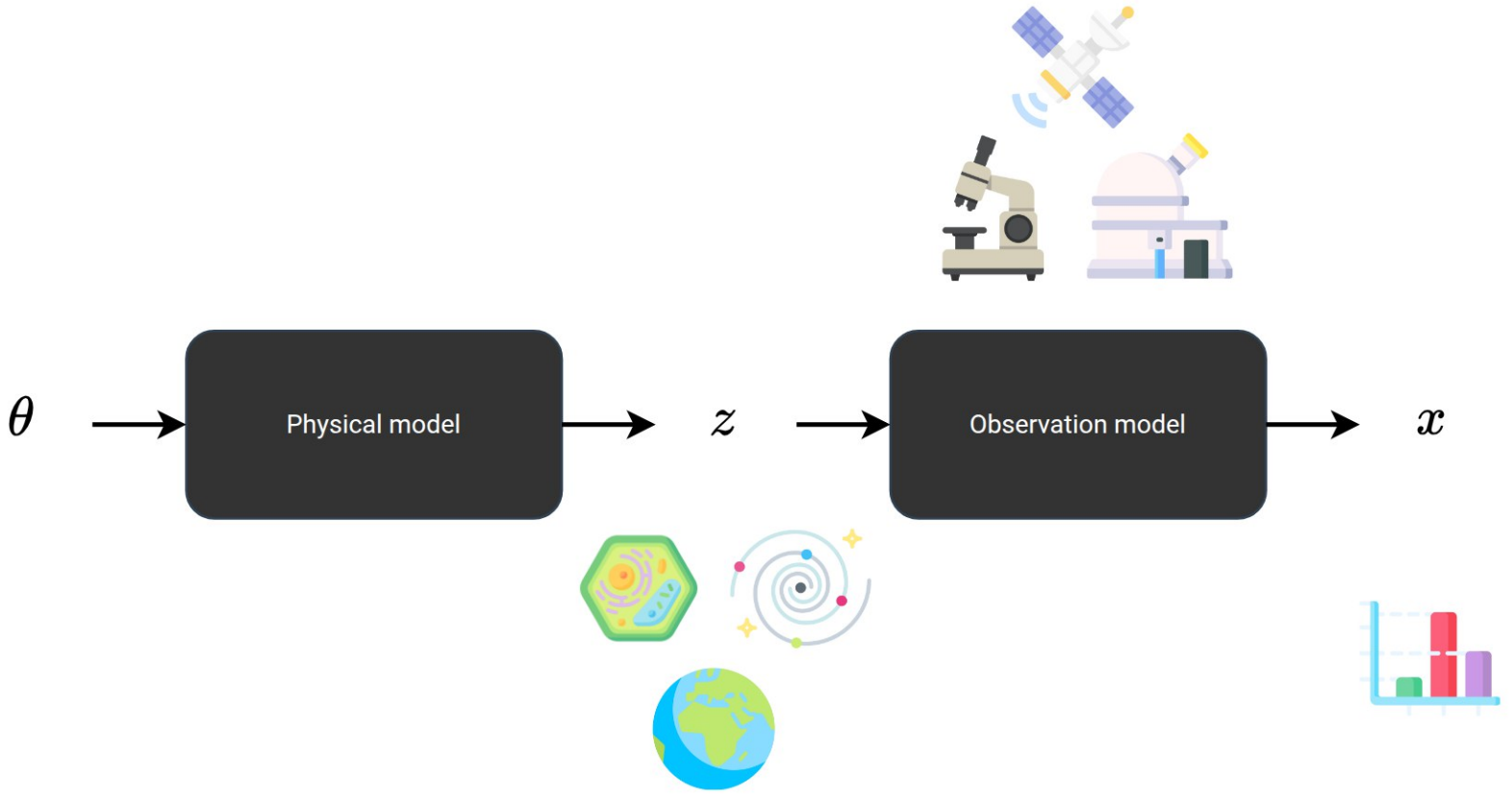
Latent states: the atmosphere z, its temperature profile, its molecules, its clouds.

VHS 1256 b. Illustration: NASA, ESA, CSA, Joseph Olmsted (STScI).

Exoplanets

Parameters: the bulk properties θ, C/O, metallicity, gravity, radius, clouds, mixing. Different values, different atmospheres.

Ten hot Jupiters studied with Hubble (Sing et al, Nature 2016). Illustration: ESA/Hubble & NASA.



Inverse problems in science

Given an observation x, estimate the posterior distribution

$$p(\theta|x) = \frac{p(x|\theta)p(\theta)}{p(x)}, \quad \text{with} \quad p(x|\theta) = \int p(x|z)p(z|\theta)\mathrm{d}z,$$

of the parameters θ, with the latent states z integrated out.

Generation





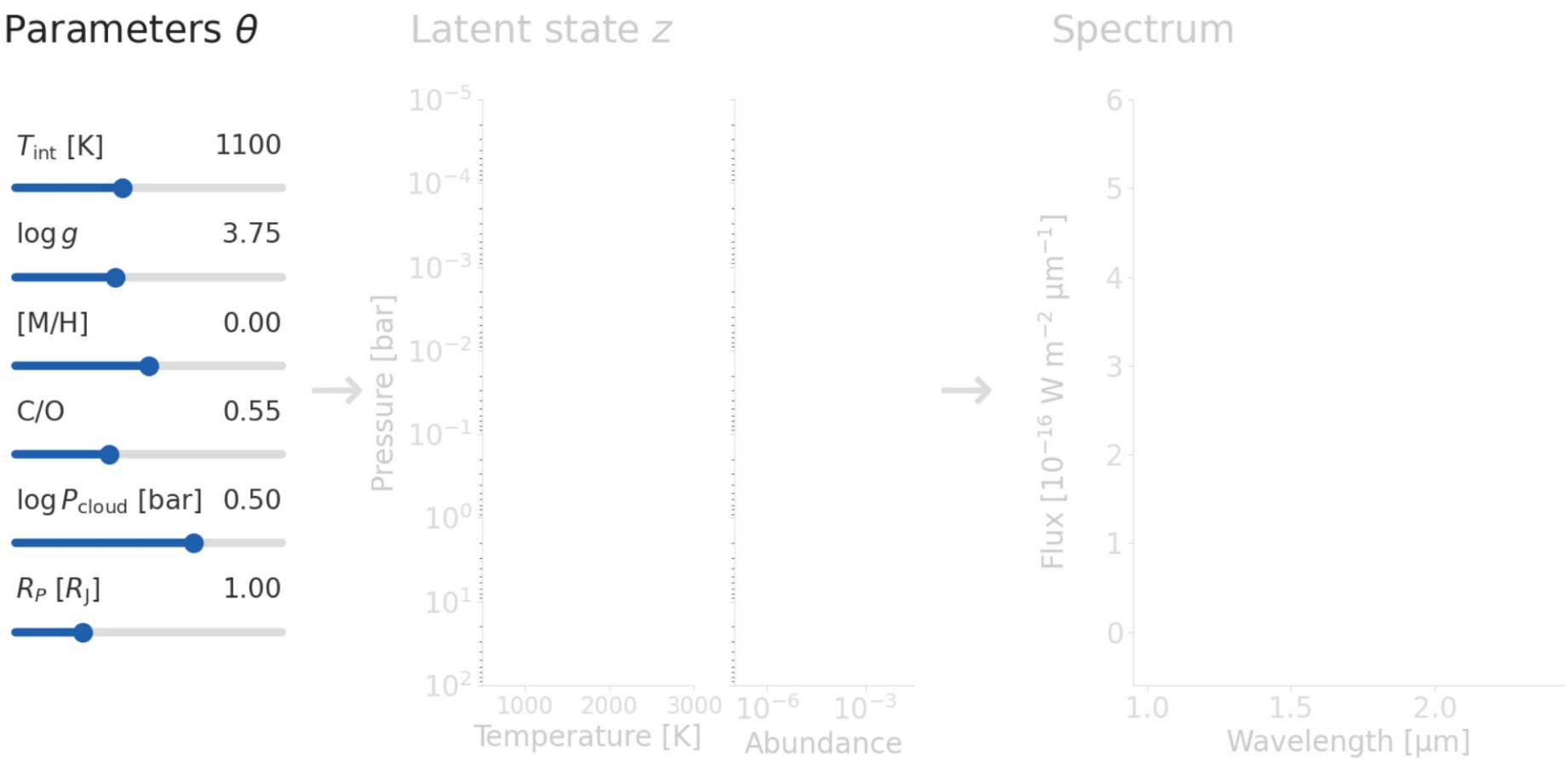


	Prior $p(\theta)$	**Physics** $p(z\|\theta)$	**Instrument** $p(x\|z)$
Ocean currents	viscosity, mixing, drag	Navier–Stokes	altimetry, noise
Gravitational waves	masses, spins, distance	general relativity	detector response, noise
Exoplanets	C/O, metallicity, clouds	radiative transfer	spectrograph, noise

Inference

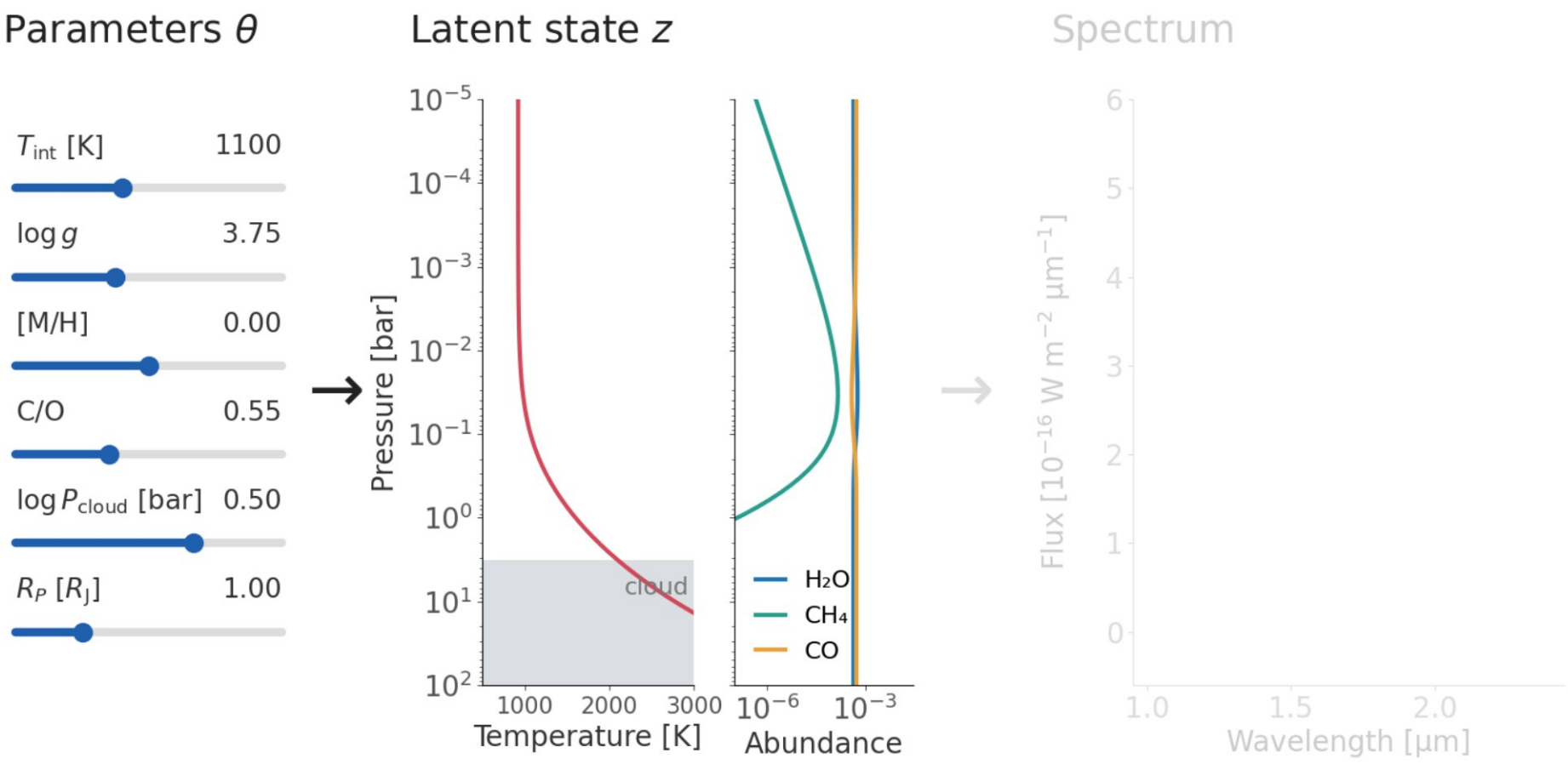
A minimal exoplanet simulator

Prior, $\theta \sim p(\theta)$: six numbers, each uniform over a range.



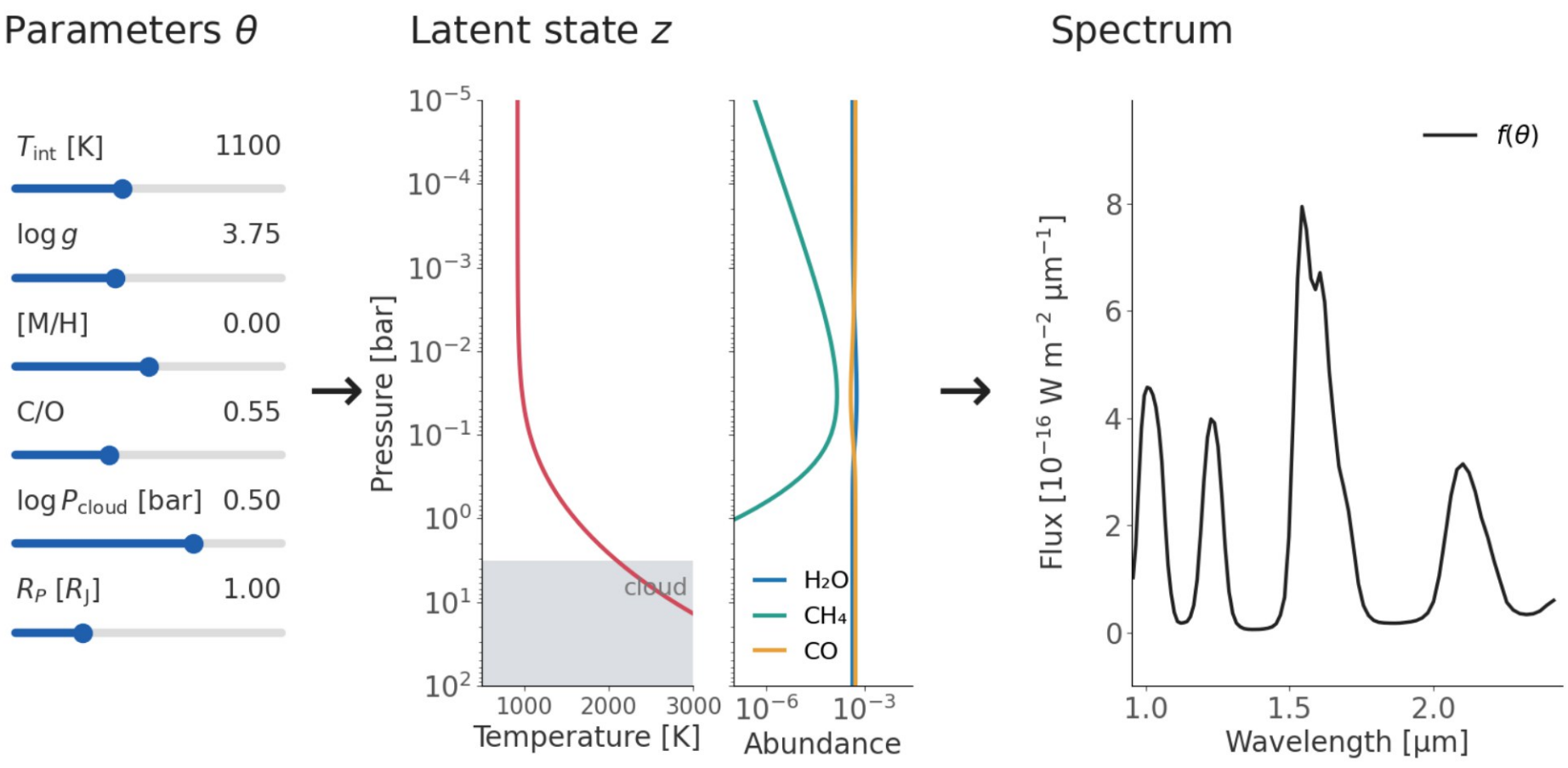
A minimal exoplanet simulator

Physics, $z \sim p(z|\theta)$: a temperature profile, chemistry, and a cloud deck.



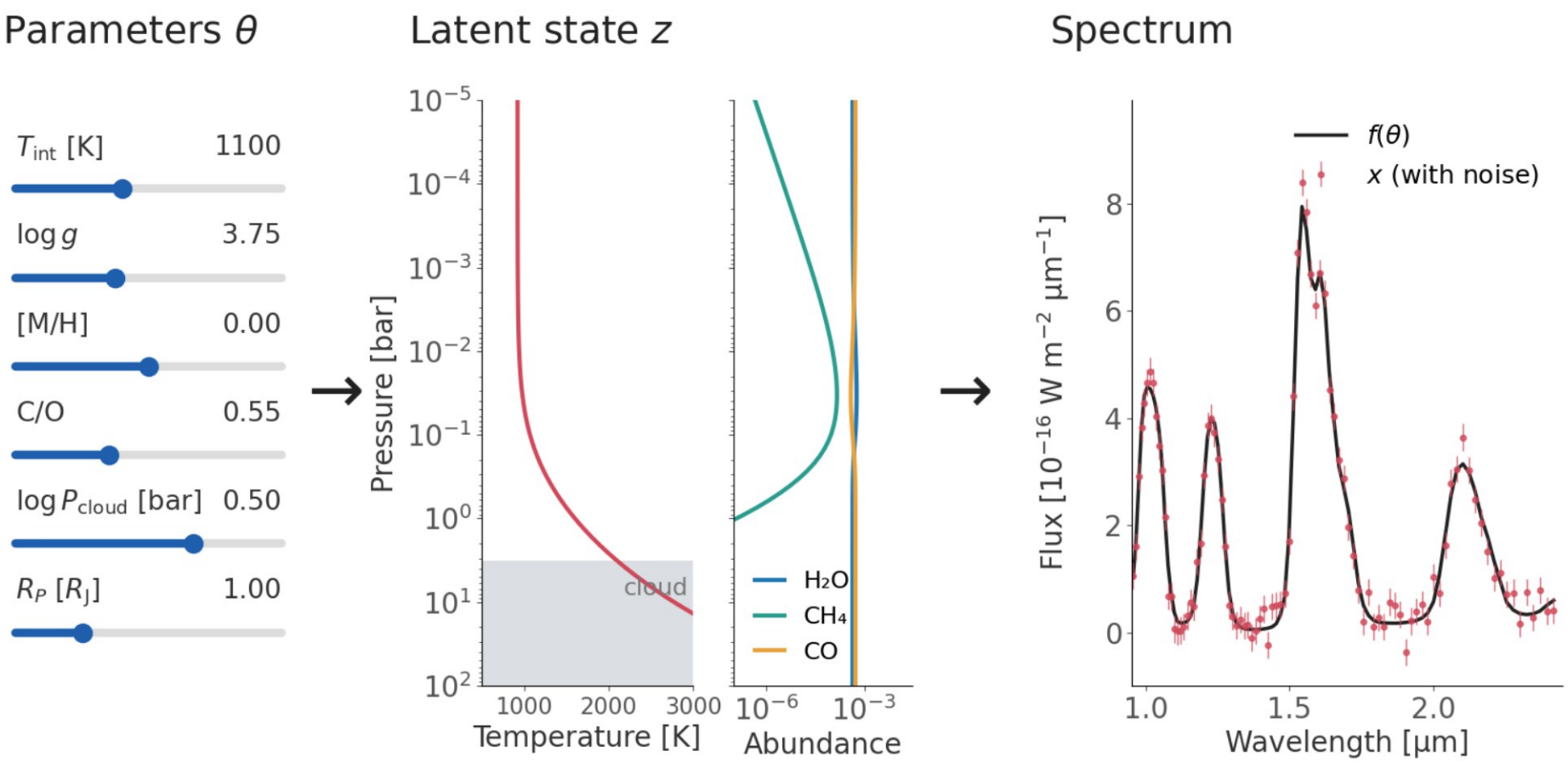
A minimal exoplanet simulator

Radiative transfer: the light $f(\theta)$ that the atmosphere emits.



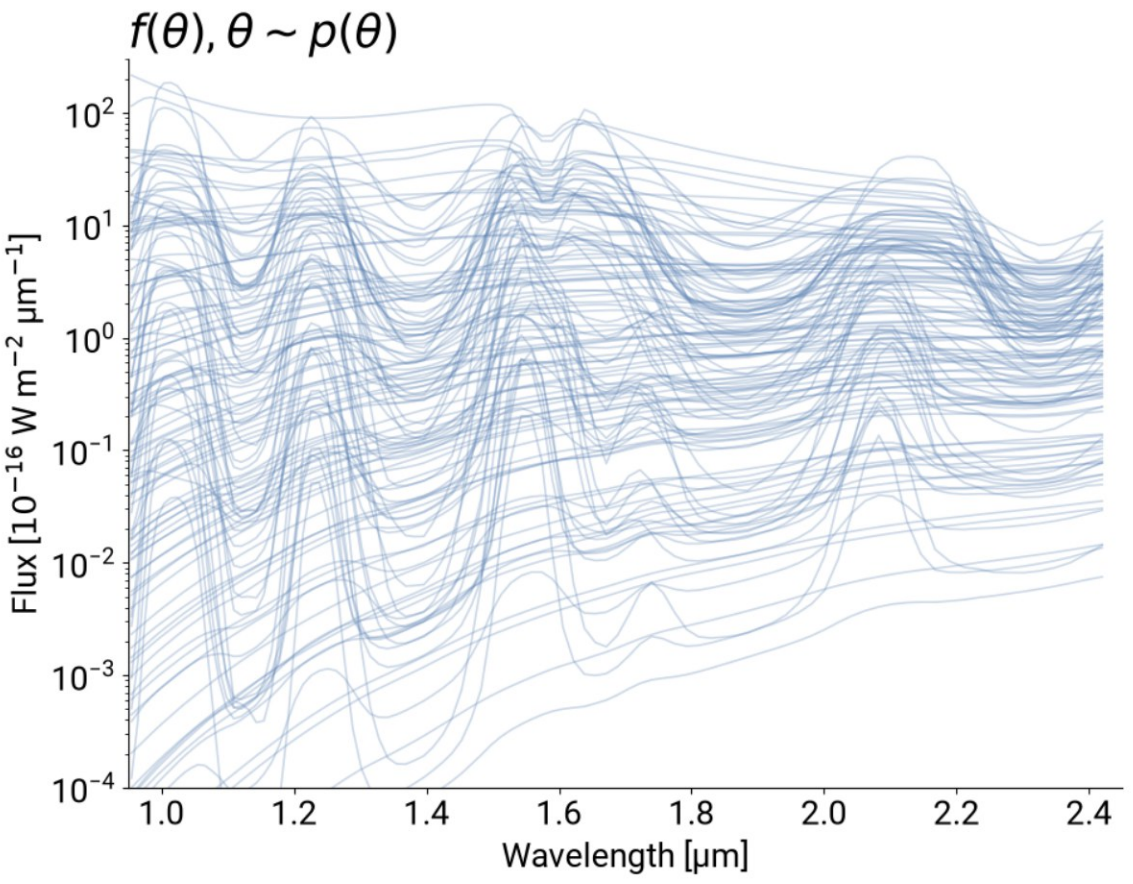
A minimal exoplanet simulator

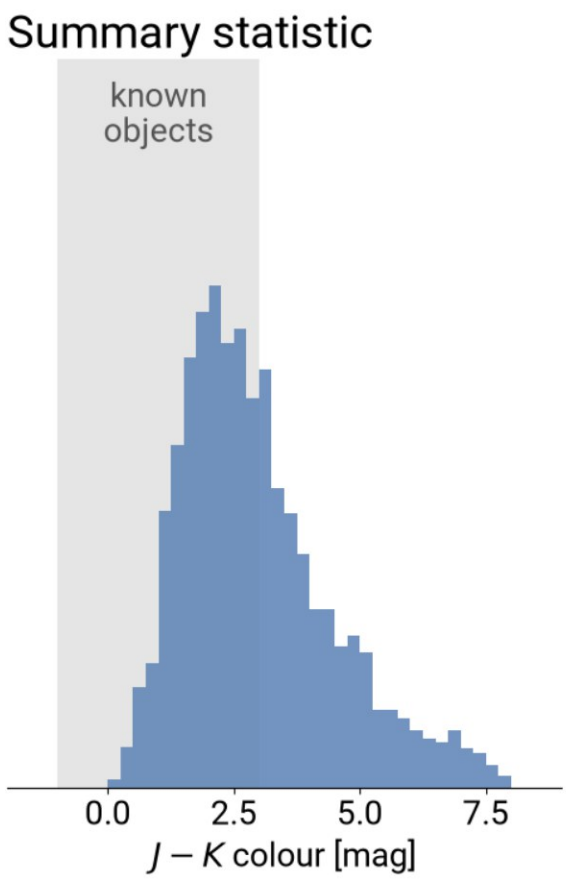
Instrument, $x \sim p(x|z)$: binned, with noise. This is all we observe.

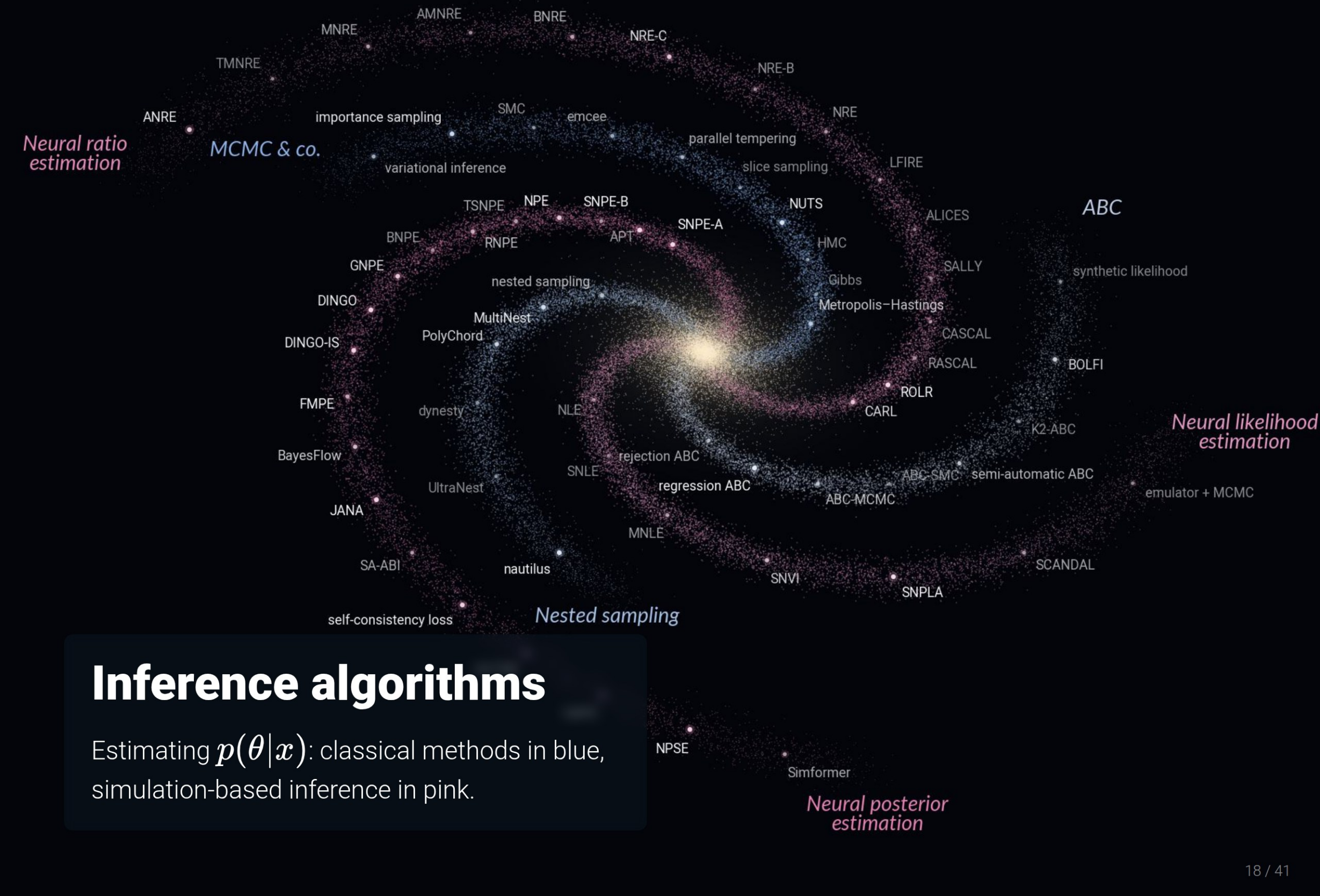


Prior predictive check

Does the **forward model** agree with what we know? Draw $\theta \sim p(\theta)$, simulate, and compare summary statistics with domain knowledge, before seeing any data.

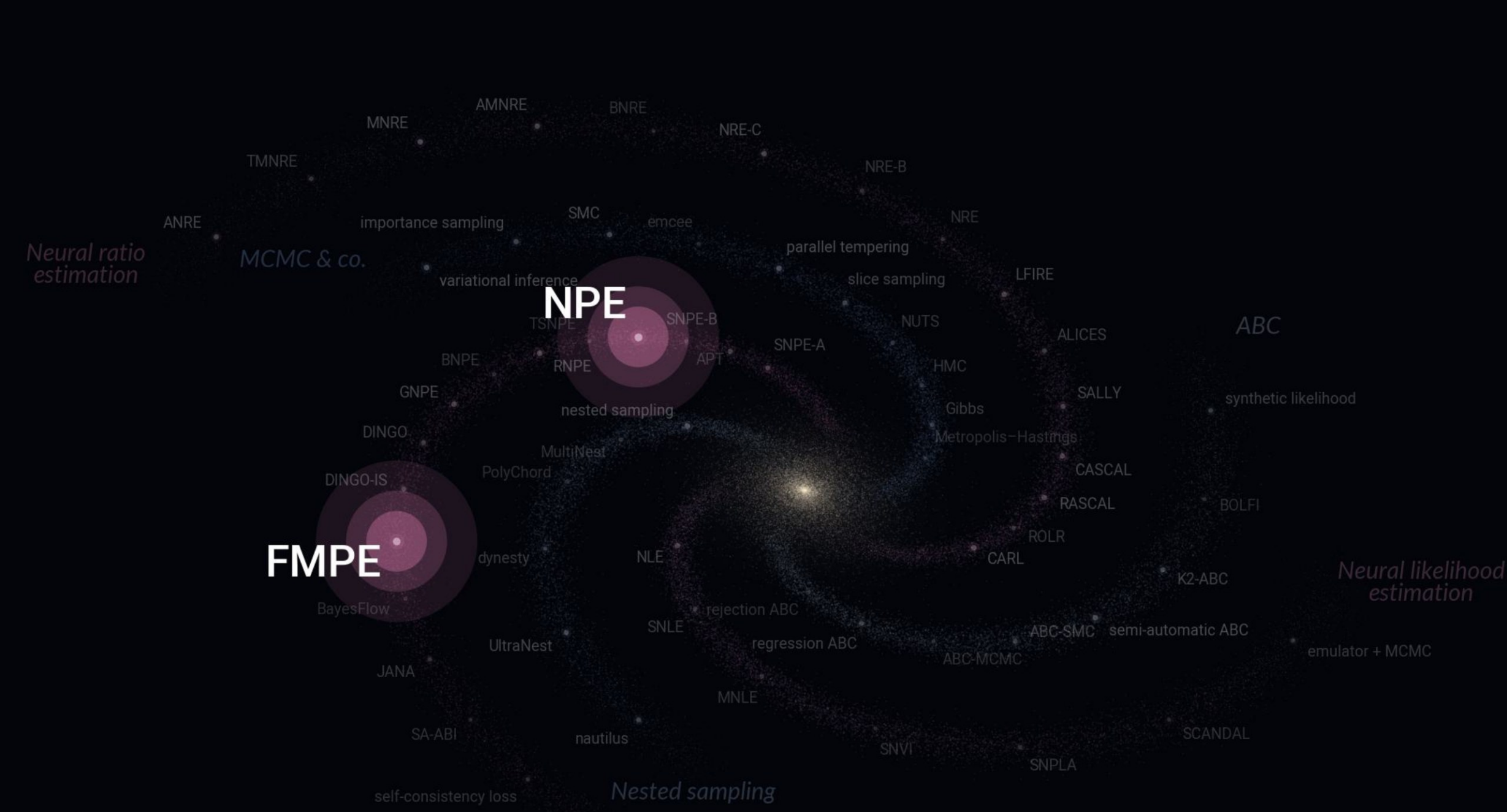






Inference algorithms

Estimating $p(\theta|x)$: classical methods in blue, simulation-based inference in pink.



Inference algorithms

Estimating $p(\theta|x)$: classical methods in blue, simulation-based inference in pink.

Neural posterior estimation

Train a conditional density estimator $q_\phi(\theta|x)$, typically a **normalizing flow**, to output the posterior of **any** observation x:

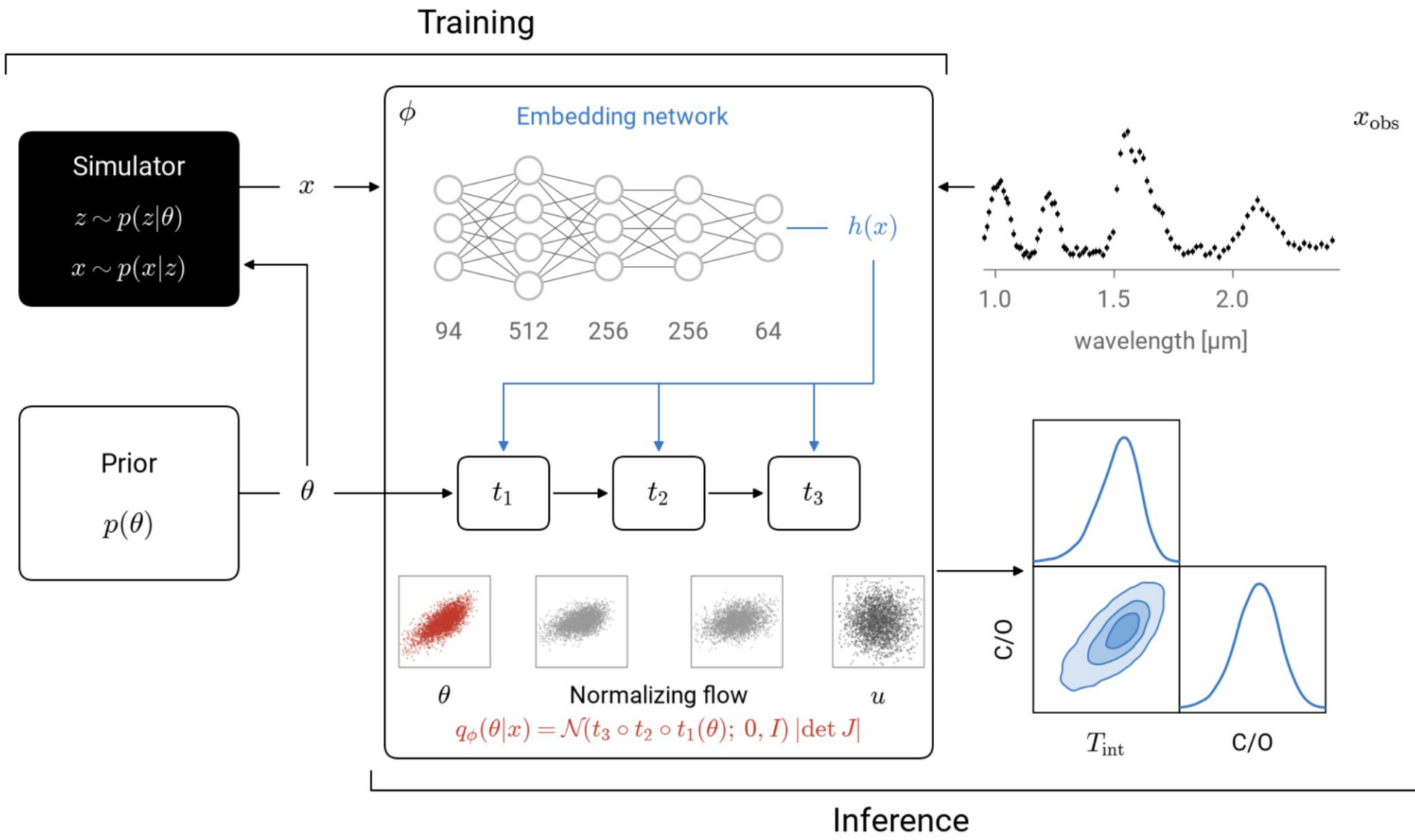
$$\phi^\star = \arg\min_\phi \; \mathbb{E}_{p(\theta,x)} \left[-\log q_\phi(\theta|x) \right].$$

The expectation is over pairs (θ, x) from the simulator: no likelihood is ever evaluated.

Up to a constant, this loss is the expected divergence to the true posterior,

$$\mathbb{E}_{p(x)} \, \mathbf{KL}\big(p(\theta|x) \,\|\, q_\phi(\theta|x)\big).$$

> With enough simulations and network capacity, the trained network is the posterior, for every observation.

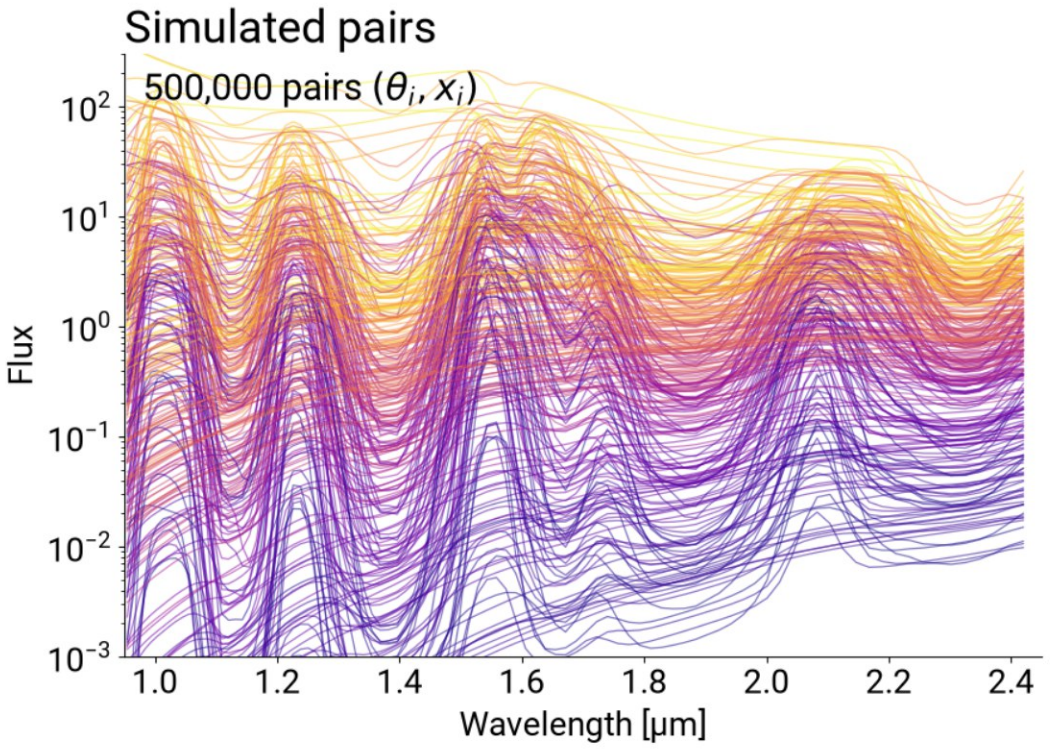


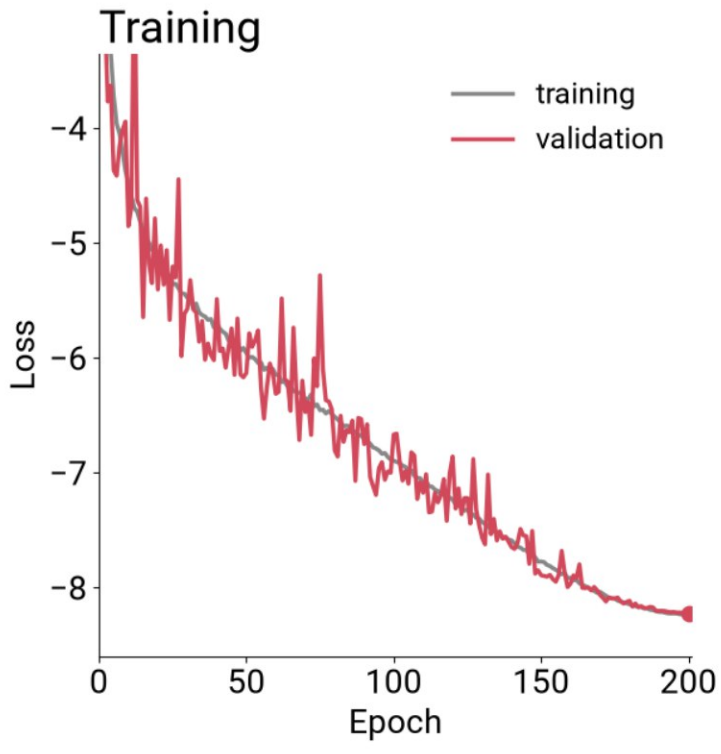
The density is exact, by the change of variables, with $u = t_3 \circ t_2 \circ t_1(\theta)$:

$$\log q_\phi(\theta|x) = \log \mathcal{N}(u; 0, I) + \sum_{k=1}^{3} \log \left|\det J_{t_k}\right| .$$

Training

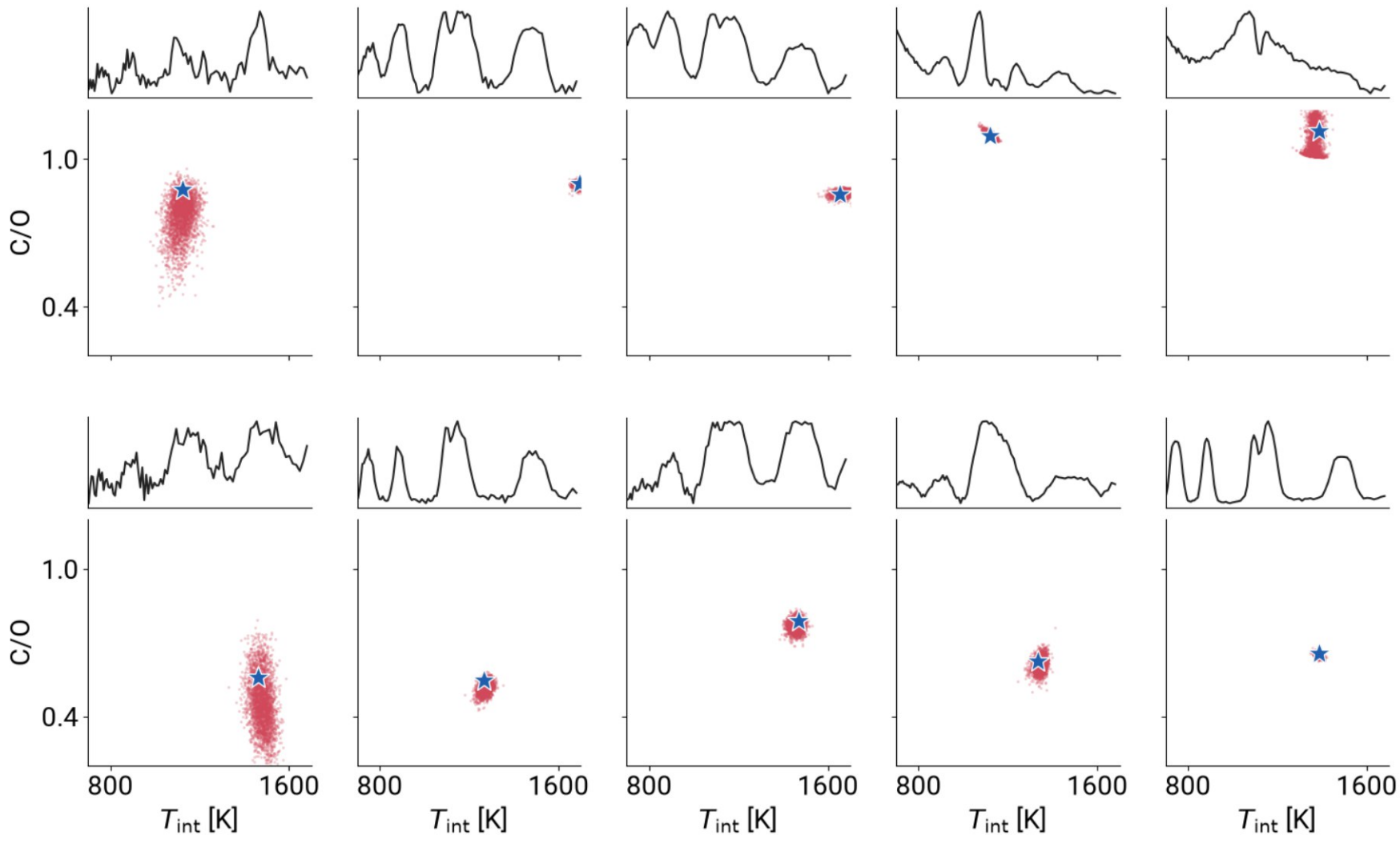
Simulate pairs (θ_i, x_i) from the prior and the simulator, then minimize the loss over them. Noise is added on the fly.





Amortization

The network is trained once. Then, any new spectrum gets its posterior in a fraction of a second.

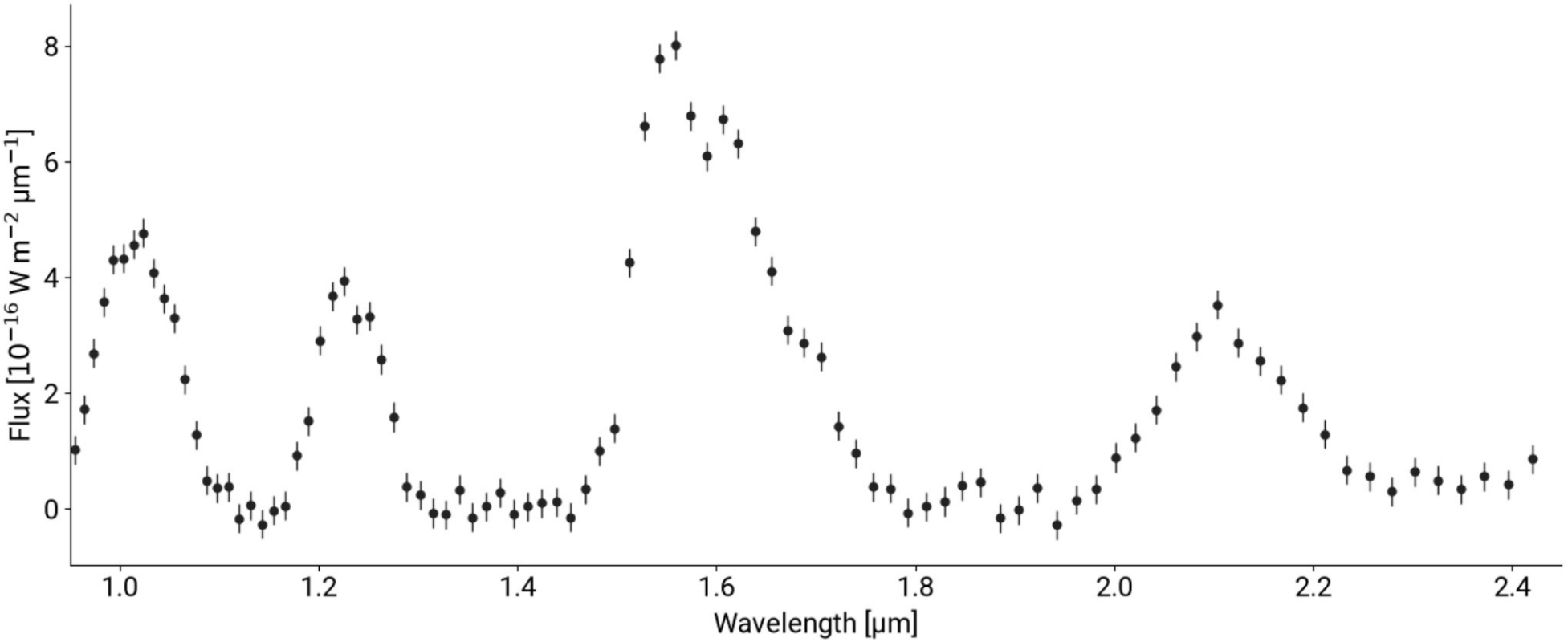


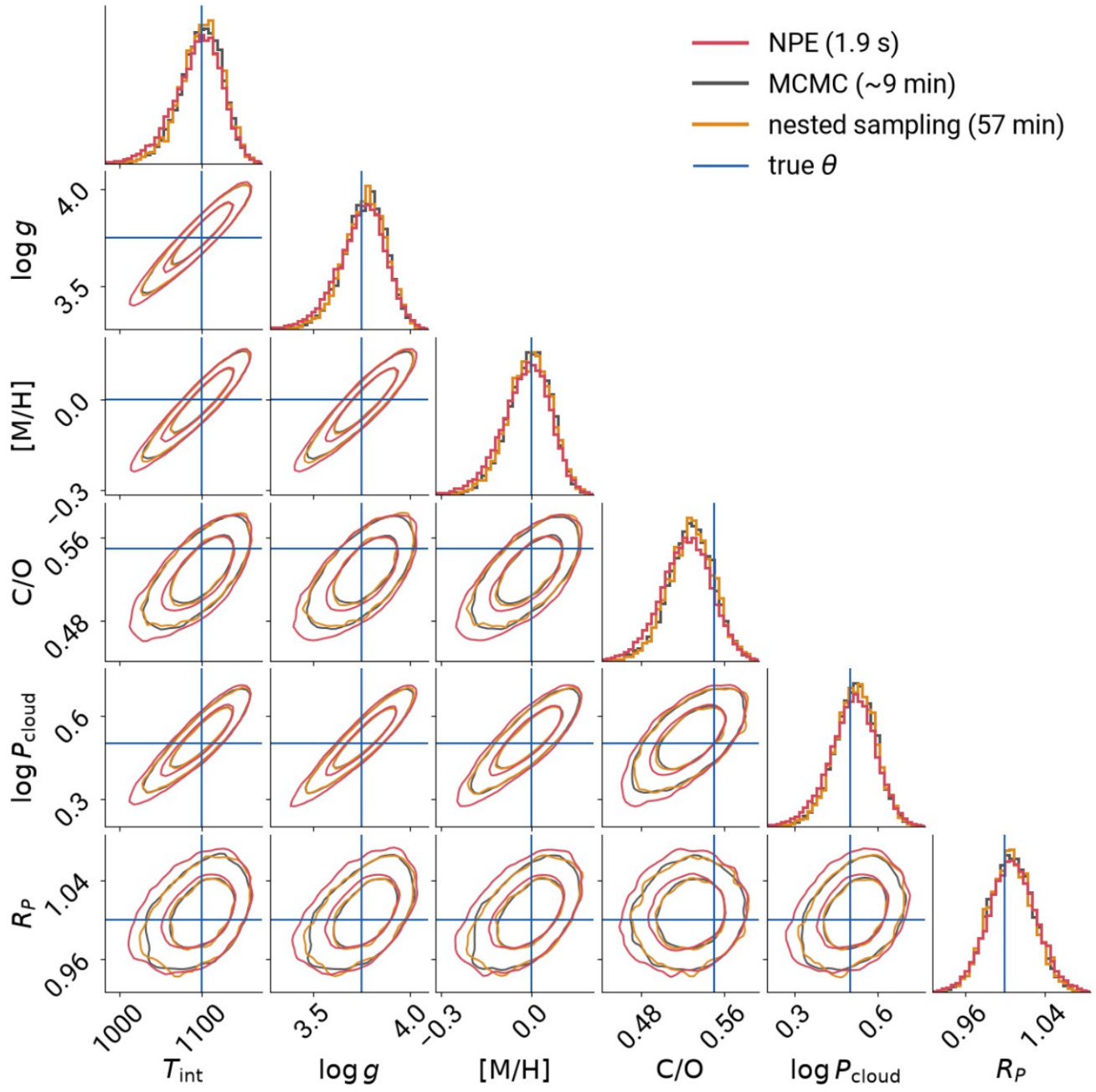
NPE: 1.9 s for 10 spectra

MCMC or nested sampling: ~90 min

Results on the toy problem

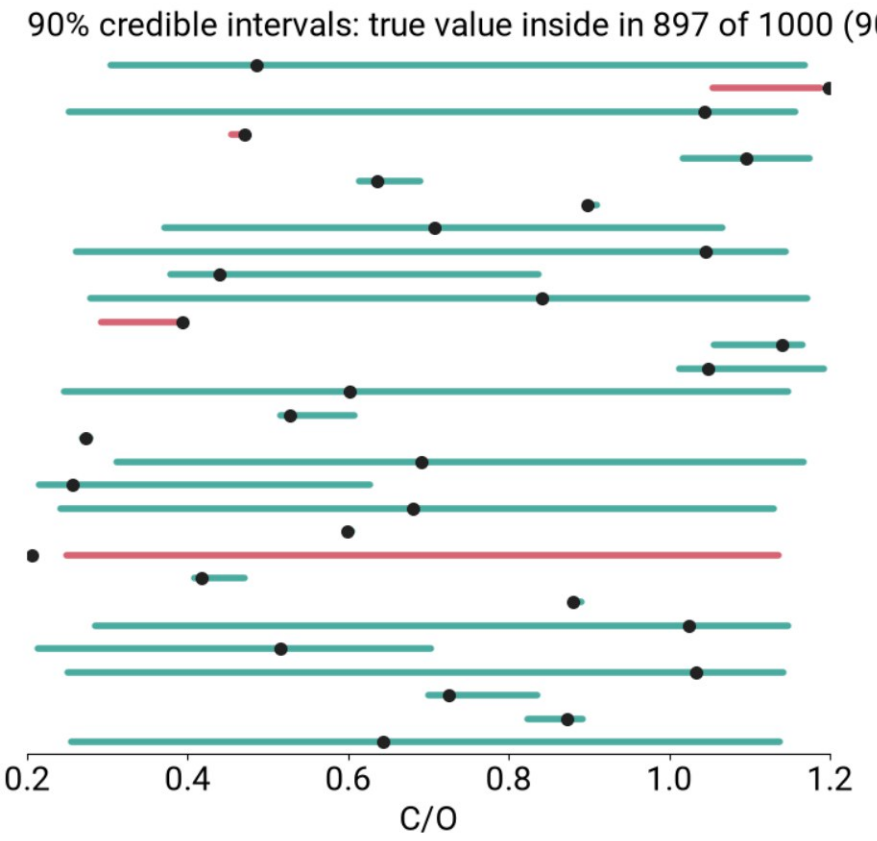
An observed spectrum x_{obs}, simulated from known parameters $\theta^{\star}$ (hidden from the methods).

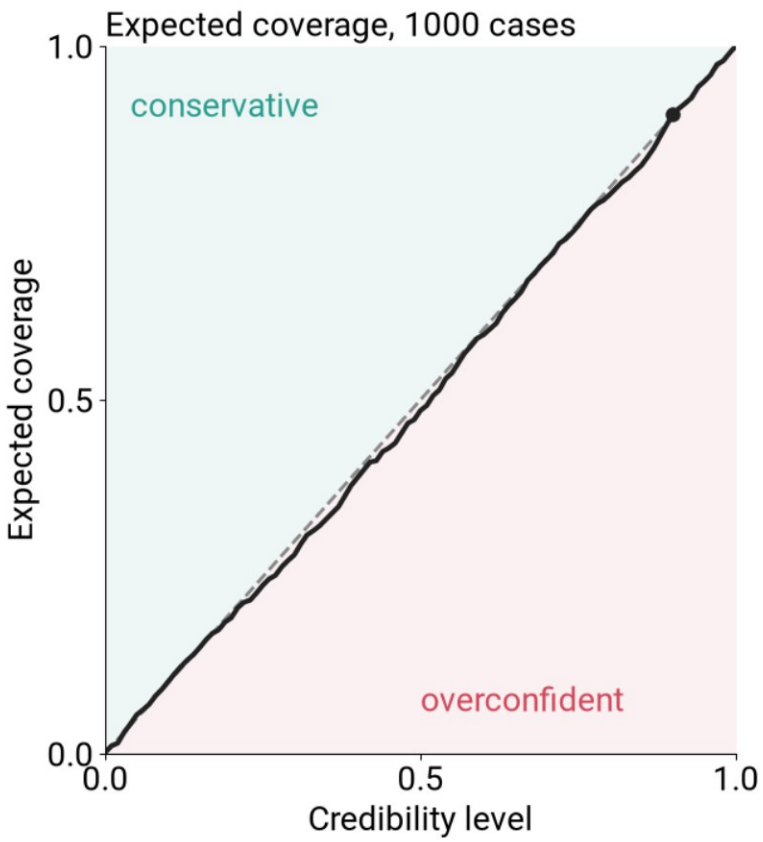



NPE (1.9 s)
MCMC (~9 min)
nested sampling (57 min)
true θ
log g
[M/H]
C/O
log P_cloud
R_P
T_int
4.0
3.5
0.0
−0.3
0.56
0.48
0.6
0.3
1.04
0.96
1000
1100

Calibration

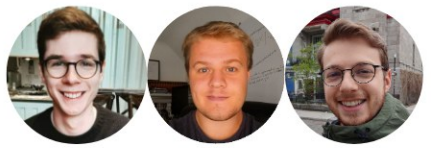
Does the **inference network** give calibrated posteriors? Check on 10^3 simulated observations, in a minute.



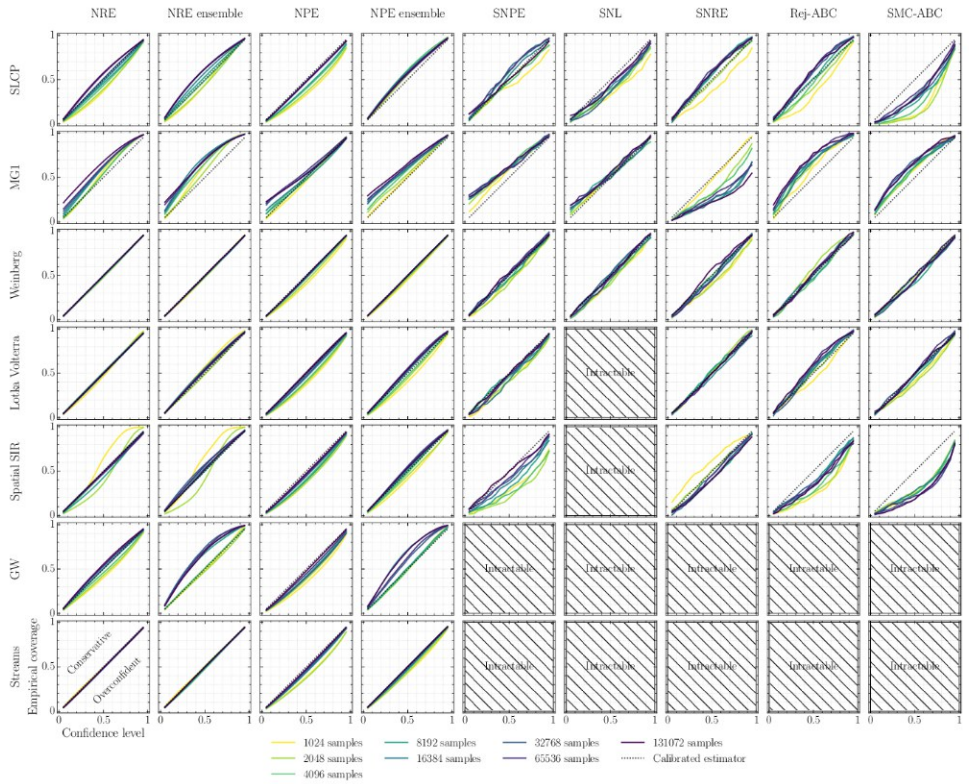


Coverage is necessary, not sufficient: the prior itself is calibrated. Stronger tests: TARP, MIRA, L-C2ST.

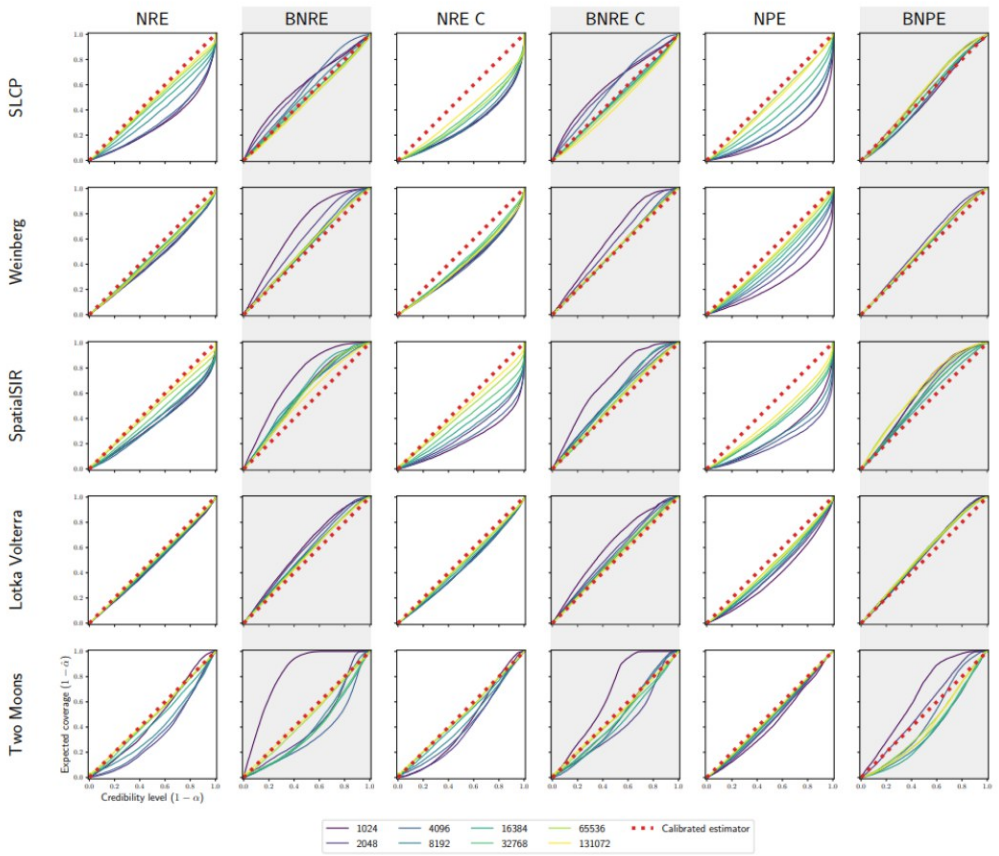
TARP: Lemos et al, ICML 2023. MIRA: Sharief et al, ICML 2026. L-C2ST: Linhart et al, NeurIPS 2023.



Neural posteriors can be **overconfident**, even when trained as recommended.
Better be overdispersed than overconfident.



Coverage of common SBI algorithms

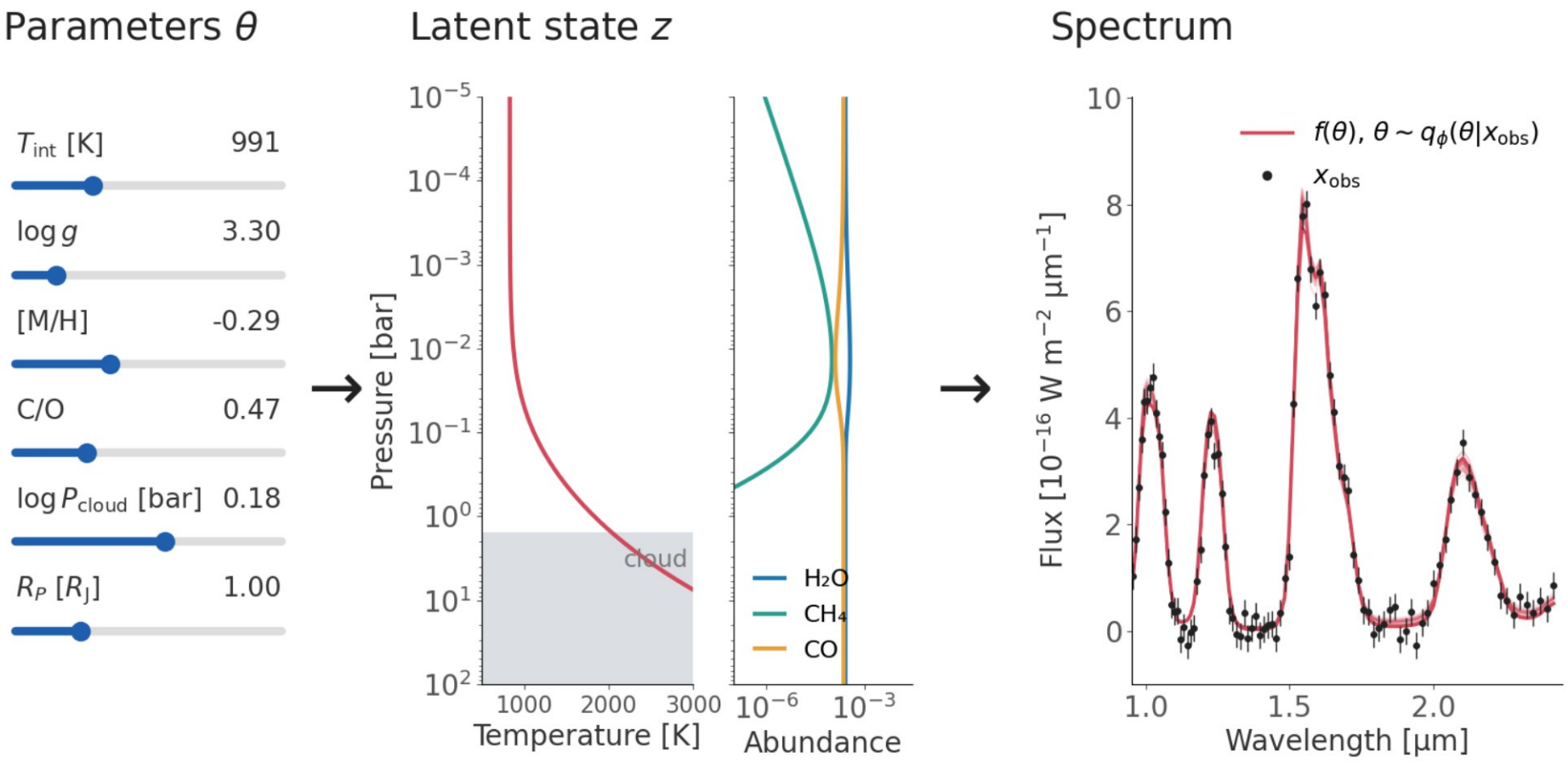


Balanced NPE (last column)

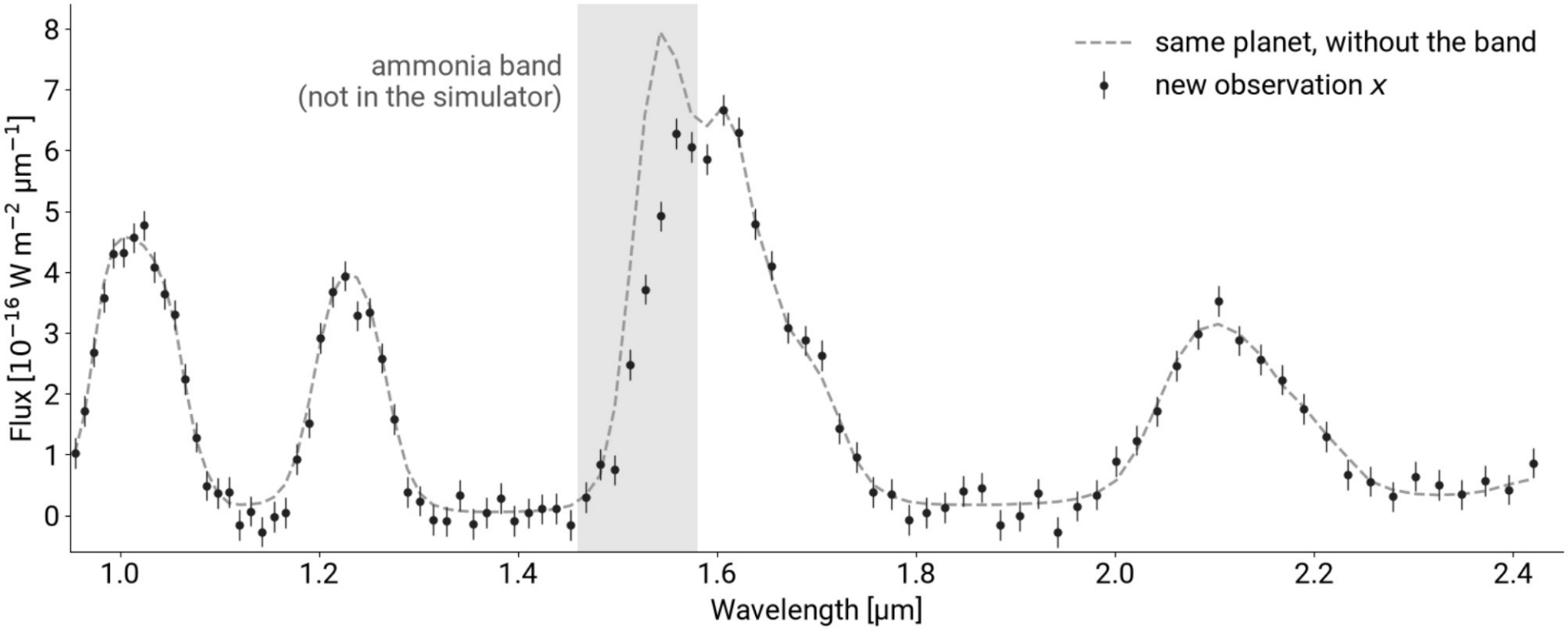
Hermans et al, TMLR 2022. Delaunoy et al, 2023. See also Falkiewicz et al, NeurIPS 2023.

Posterior predictive check

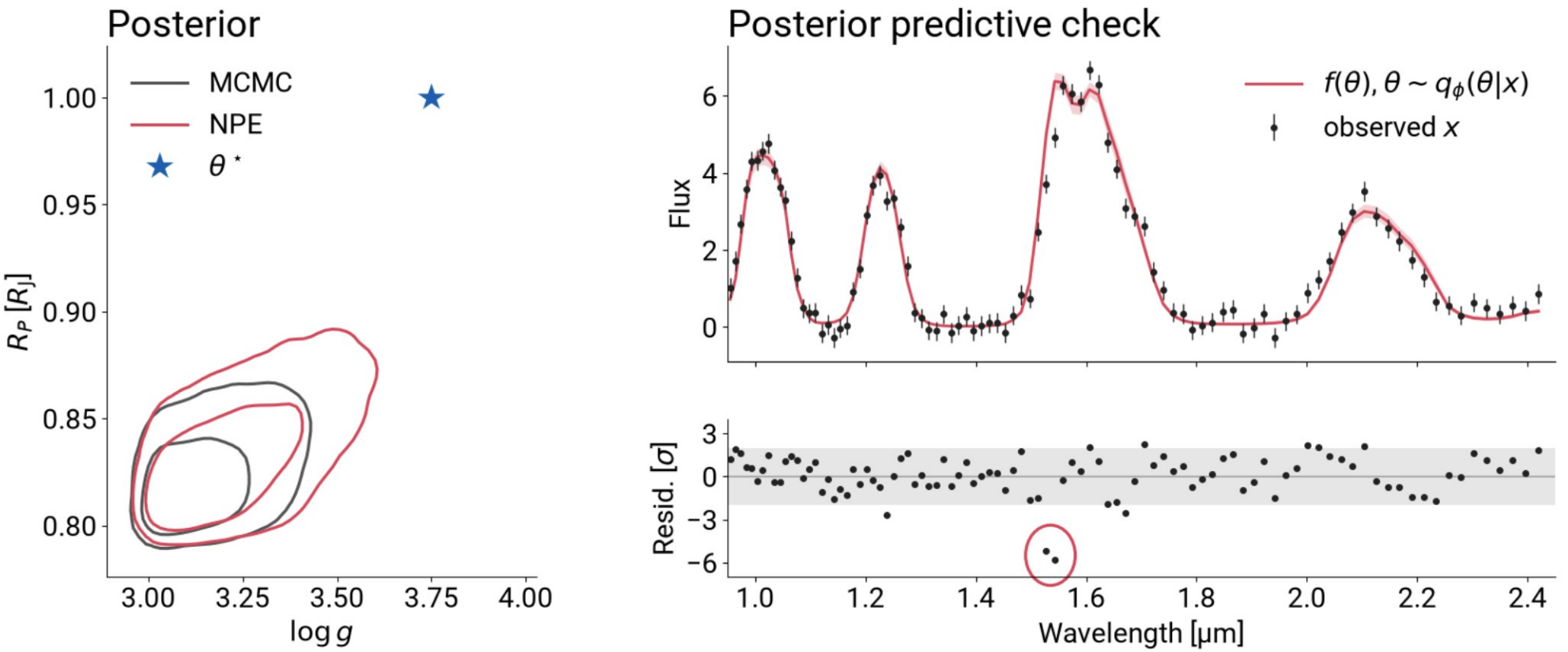
Does the **forward model** explain the observation? Draw $\theta \sim q_\phi(\theta|x_{\text{obs}})$, simulate, compare.



When the model is wrong



A new observation of the same planet, with an absorption band at 1.52 μm that the simulator does not have.



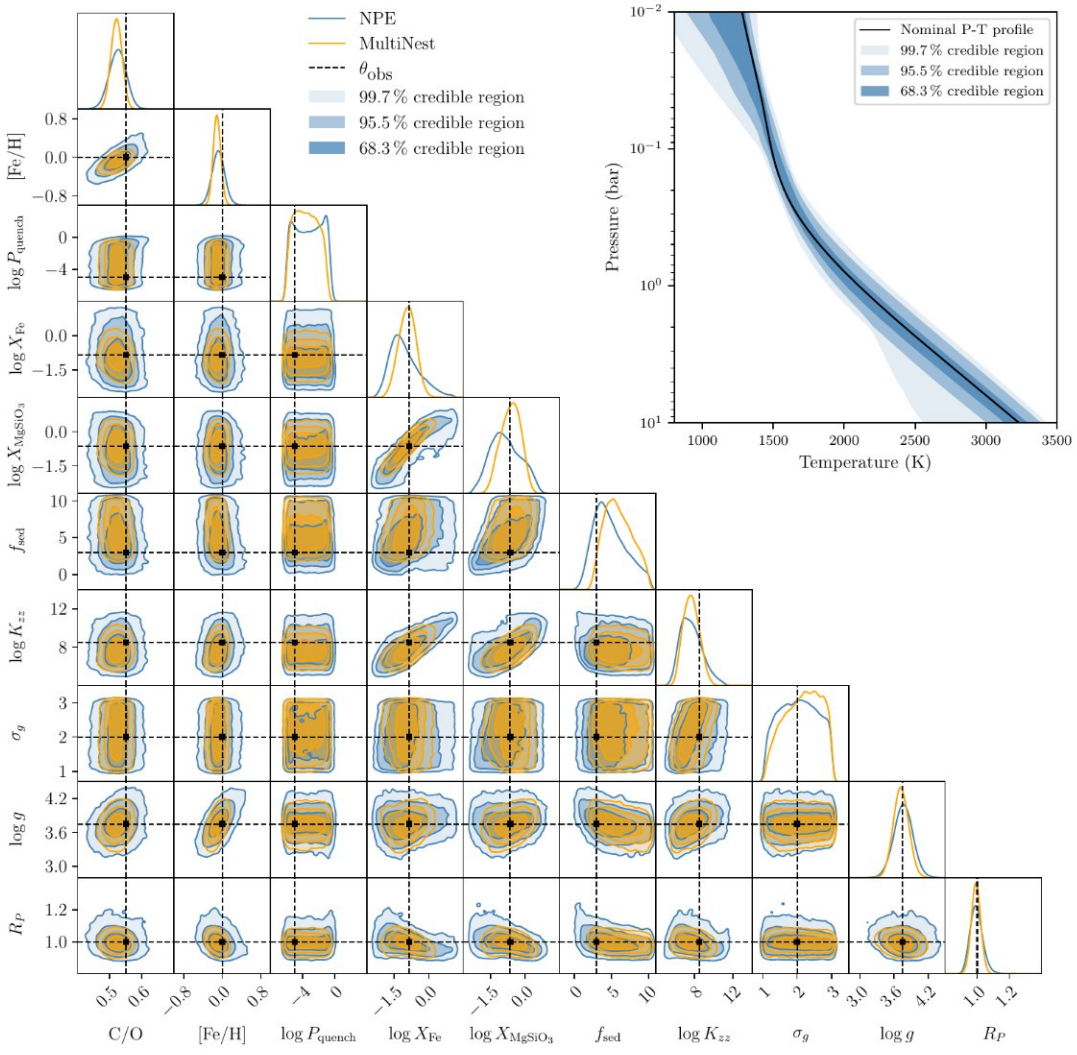
Both posteriors are **confident and wrong**.

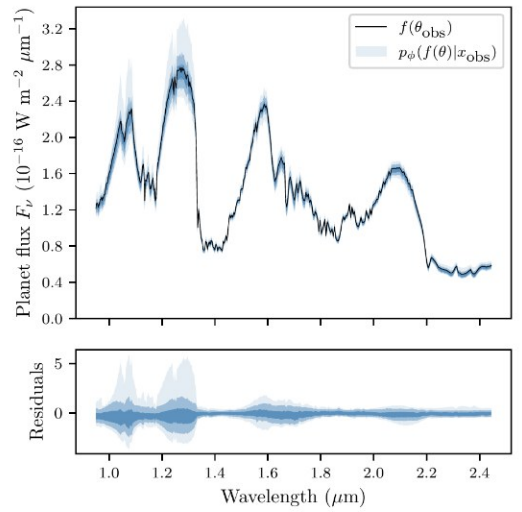
Detecting and handling misspecification: Schmitt et al, 2023; Ward et al, NeurIPS 2022.

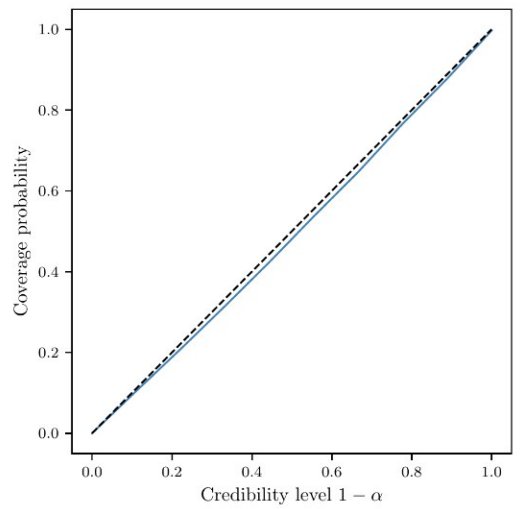


Neural posterior estimation for exoplanetary atmospheric retrieval

Malavika Vasist[1,2,⋆], François Rozet[2,⋆⋆], Olivier Absil[1,⋆⋆⋆], Paul Mollière[3], Evert Nasedkin[3], and Gilles Louppe[2]



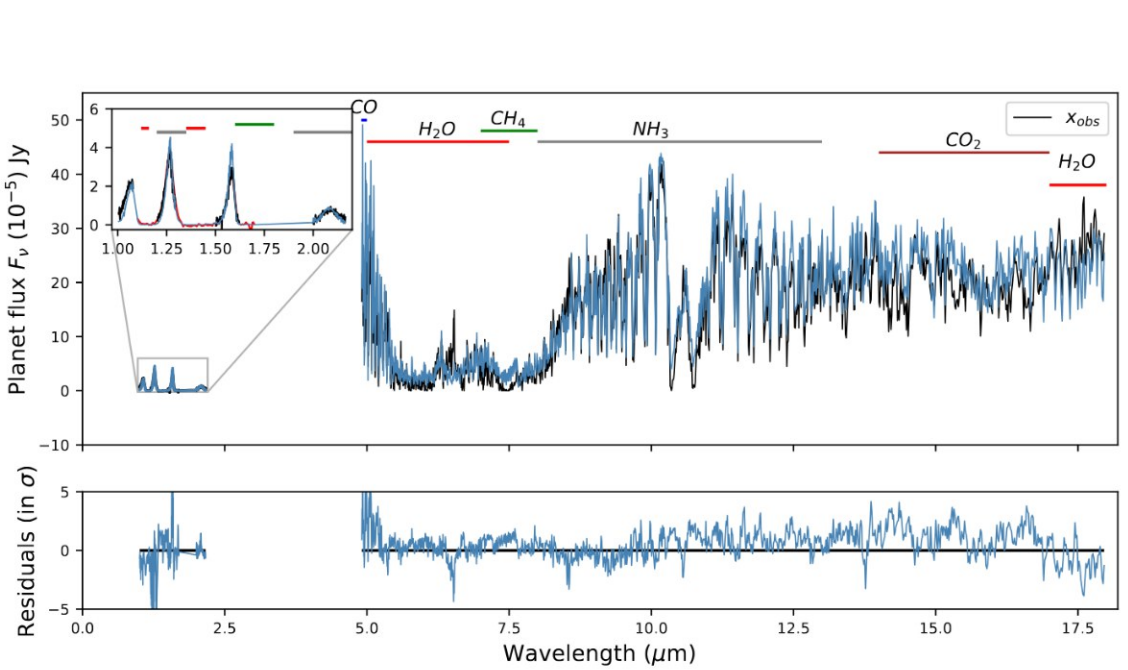


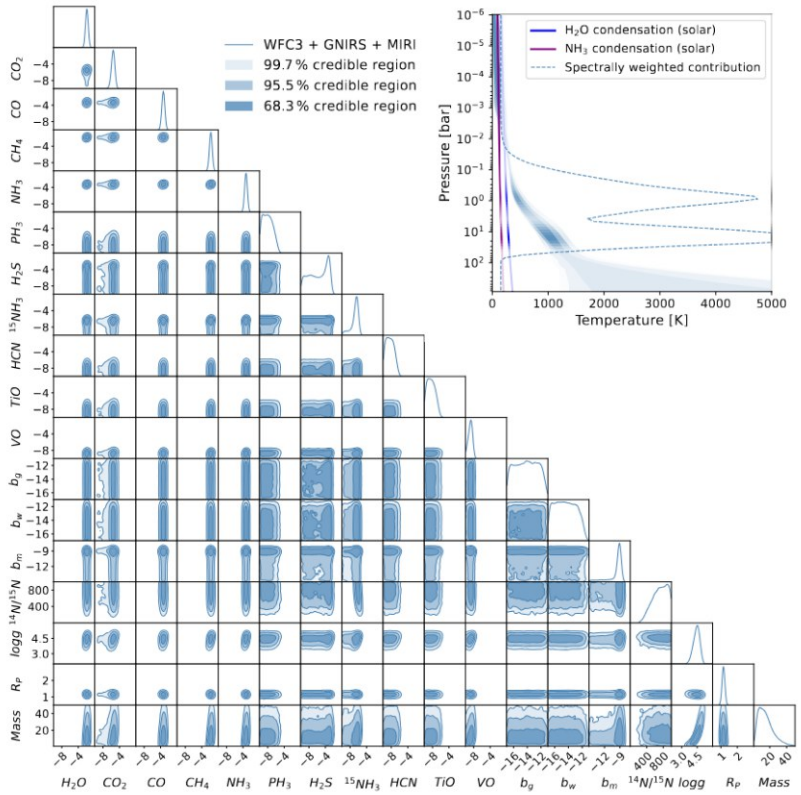




Panchromatic characterization of the Y0 brown dwarf WISEP J173835.52+273258.9 using JWST/MIRI

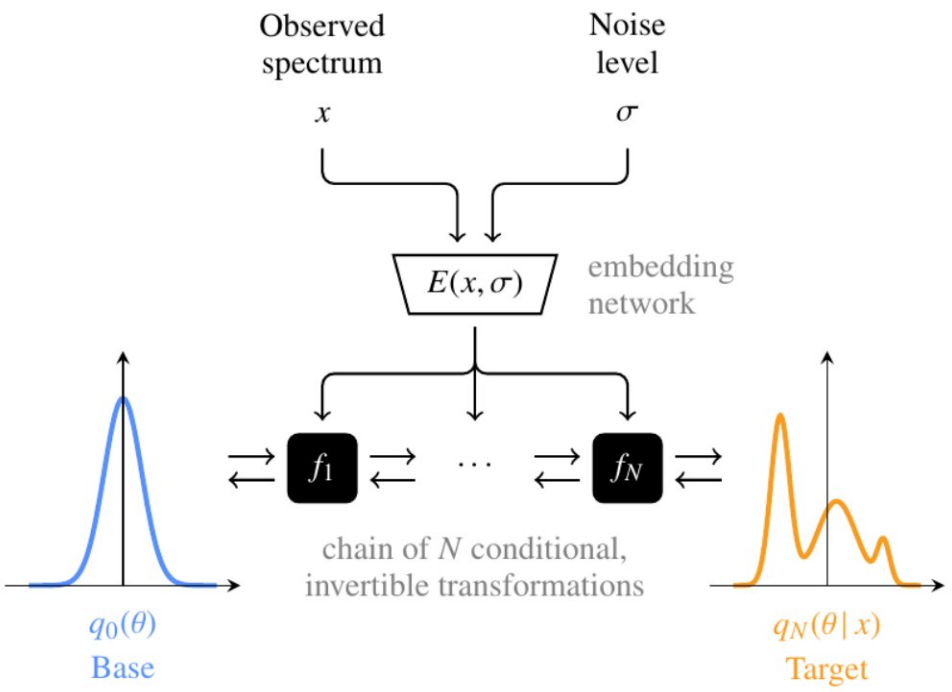
M. Vasist[1,2], P. Mollière[3], H. Kühnle[4], O. Absil[1], G. Louppe[2], Rens Waters[5,6,7], Manuel Güdel[8,4], Thomas Henning[3] David Barrado[9], Leen Decin[10], John Pye[11], and Pascal Tremblin [12]



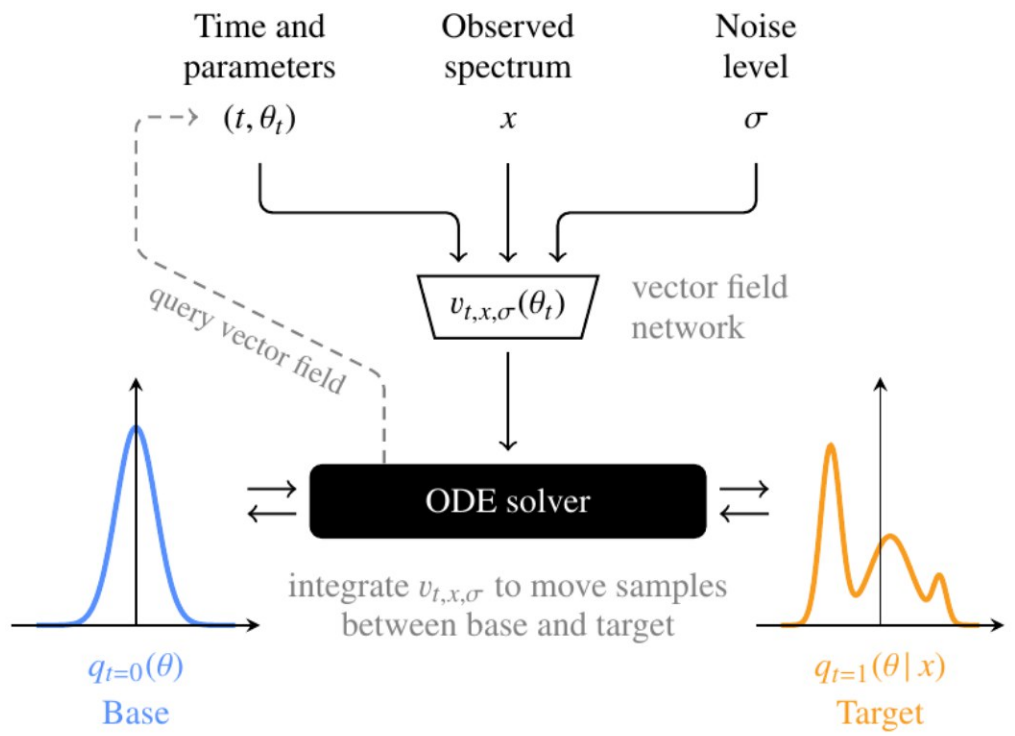


Beyond normalizing flows

NPE only needs a conditional density estimator $q_\phi(\theta|x)$. A chain of discrete transformations is one choice; a continuous flow, defined by a vector field and an ODE, is another.



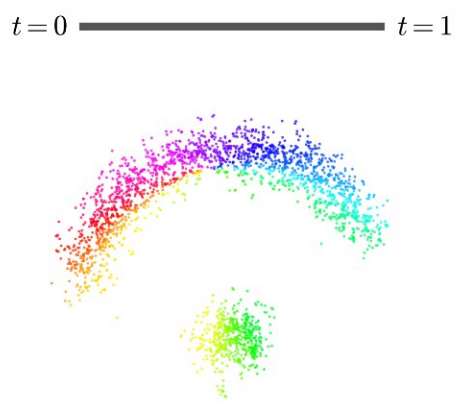
Discrete flow (NPE)



Continuous flow (FMPE)

Credits: Gebhard et al, A&A 2025.

Flow matching posterior estimation (FMPE)



Noise $\theta_0 \sim \mathcal{N}(0, I)$ flows along a vector field $v_\phi(\theta_t, t; x)$, from $t = 0$ to 1. The density is exact, by the continuous change of variables:

$$\begin{aligned}\log q_\phi(\theta|x) &= \log \mathcal{N}(\theta_0; 0, I) - \int_0^1 \operatorname{tr}\left(\frac{\partial v_\phi(\theta_t, t; x)}{\partial \theta_t}\right) dt \\ &= \log \mathcal{N}(\theta_0; 0, I) - \int_0^1 \nabla \cdot v_\phi(\theta_t, t; x)\, dt.\end{aligned}$$

Train the vector field by regression, along straight paths $\theta_t = (1 - t)\,\theta_0 + t\,\theta$:

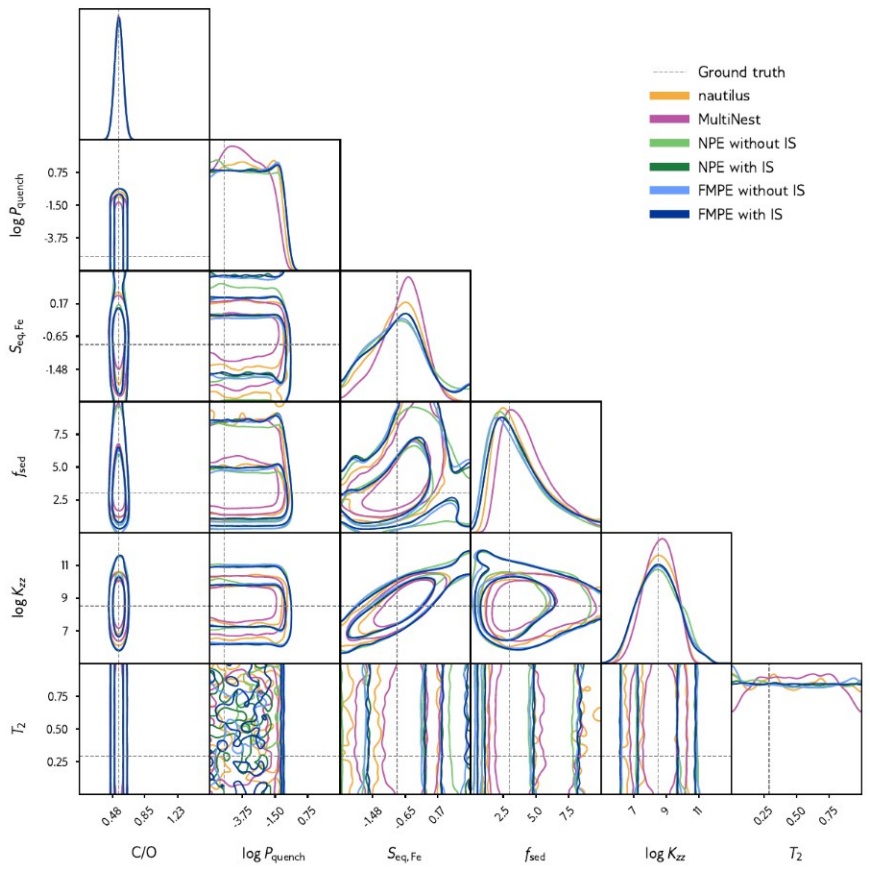
$$\min_\phi \; \mathbb{E}_{t\sim\mathcal{U}(0,1),\; \theta_0\sim\mathcal{N}(0,I),\; (\theta,x)\sim p(\theta,x)} \left[\left\| v_\phi(\theta_t, t; x) - (\theta - \theta_0)\right\|^2\right].$$

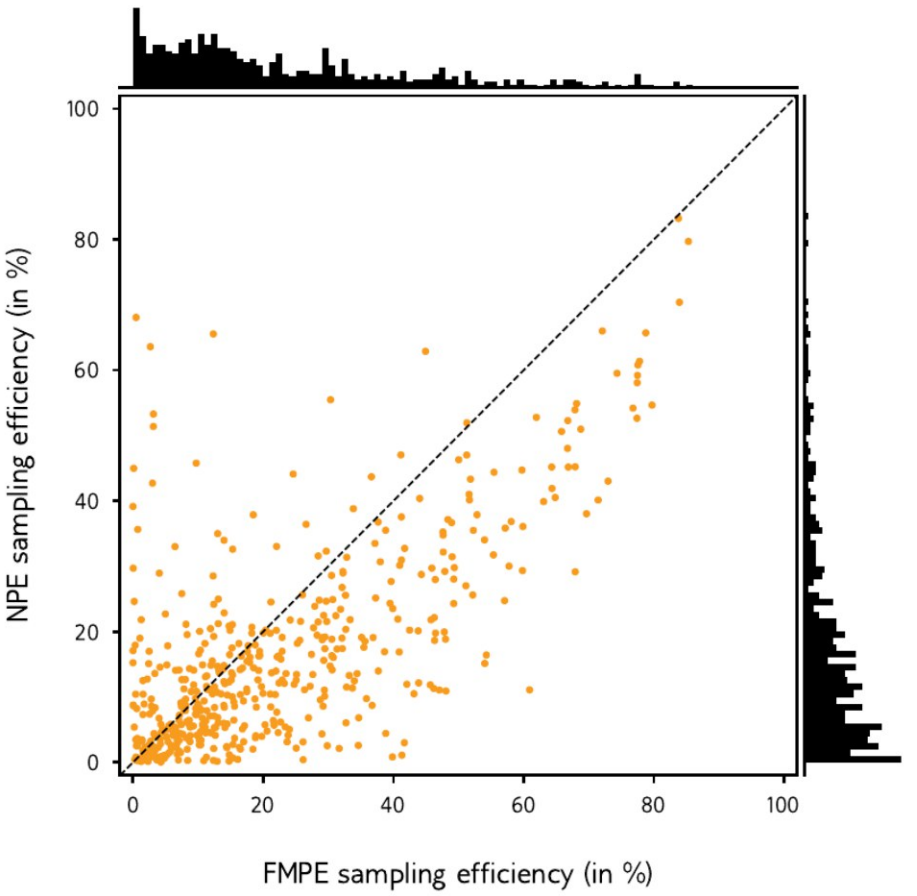
Dax et al, NeurIPS 2023. Animation: the exact flow-matching vector field towards a two-mode target.



Flow Matching for Atmospheric Retrieval of Exoplanets: Where Reliability meets Adaptive Noise Levels

Timothy D. Gebhard[1,2,*], Jonas Wildberger[1,3], Maximilian Dax[1], Annalena Kofler[1,4], Daniel Angerhausen[2], Sascha P. Quanz[2,5], and Bernhard Schölkopf[1,6]

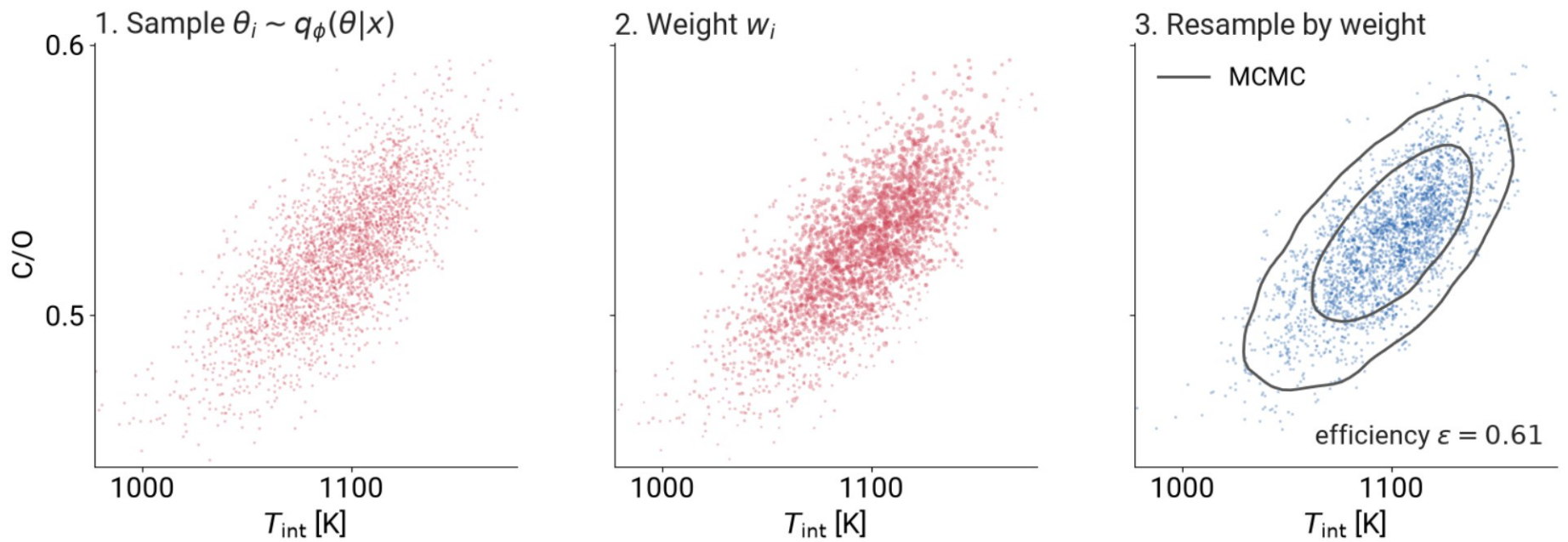




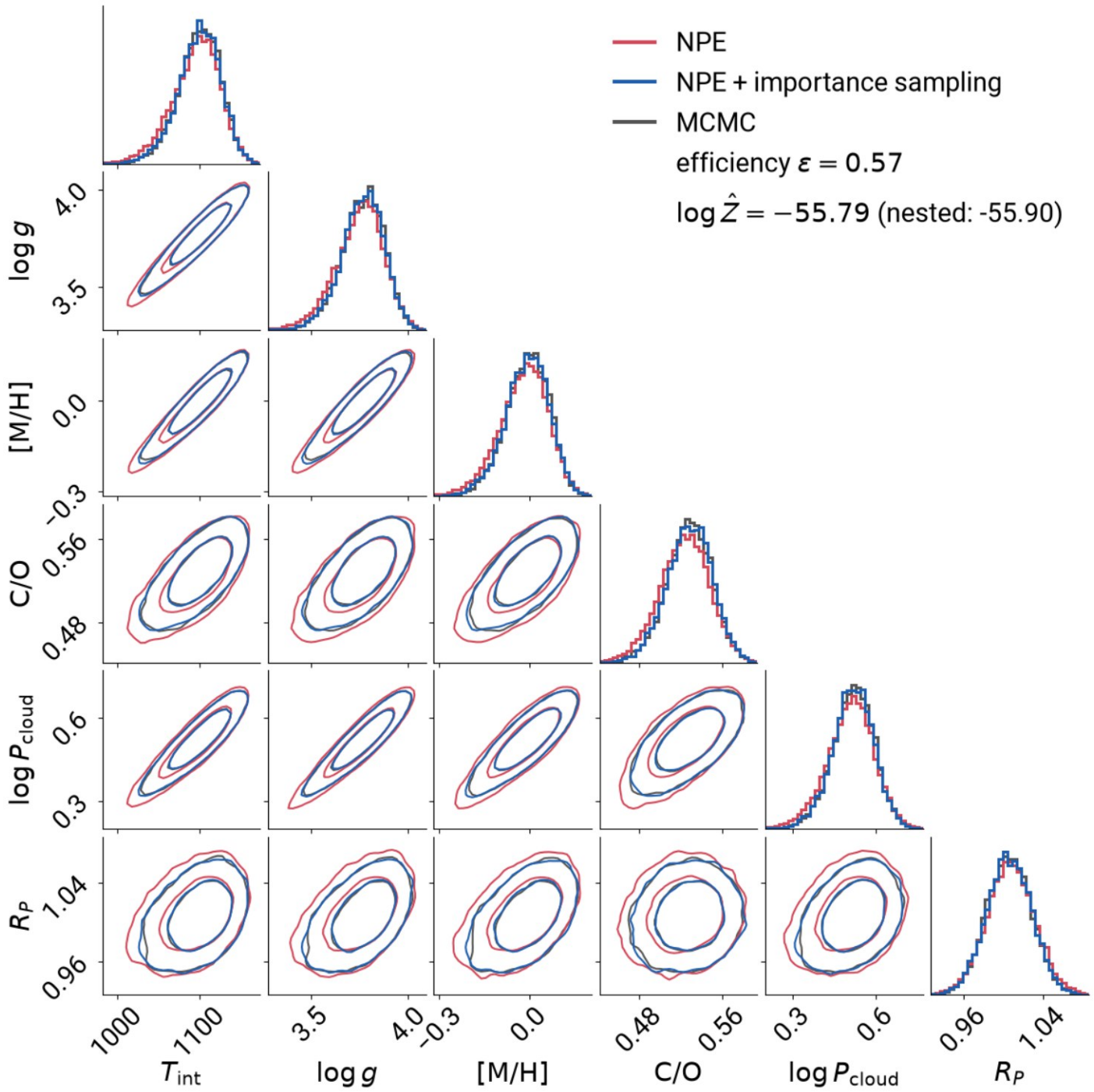
Cheat code: importance sampling

When the likelihood can be evaluated, the network becomes a **proposal**. Draw $\theta_i \sim q_\phi(\theta|x)$, then weight each sample by

$$w_i = \frac{p(x|\theta_i)\,p(\theta_i)}{q_\phi(\theta_i|x)}.$$



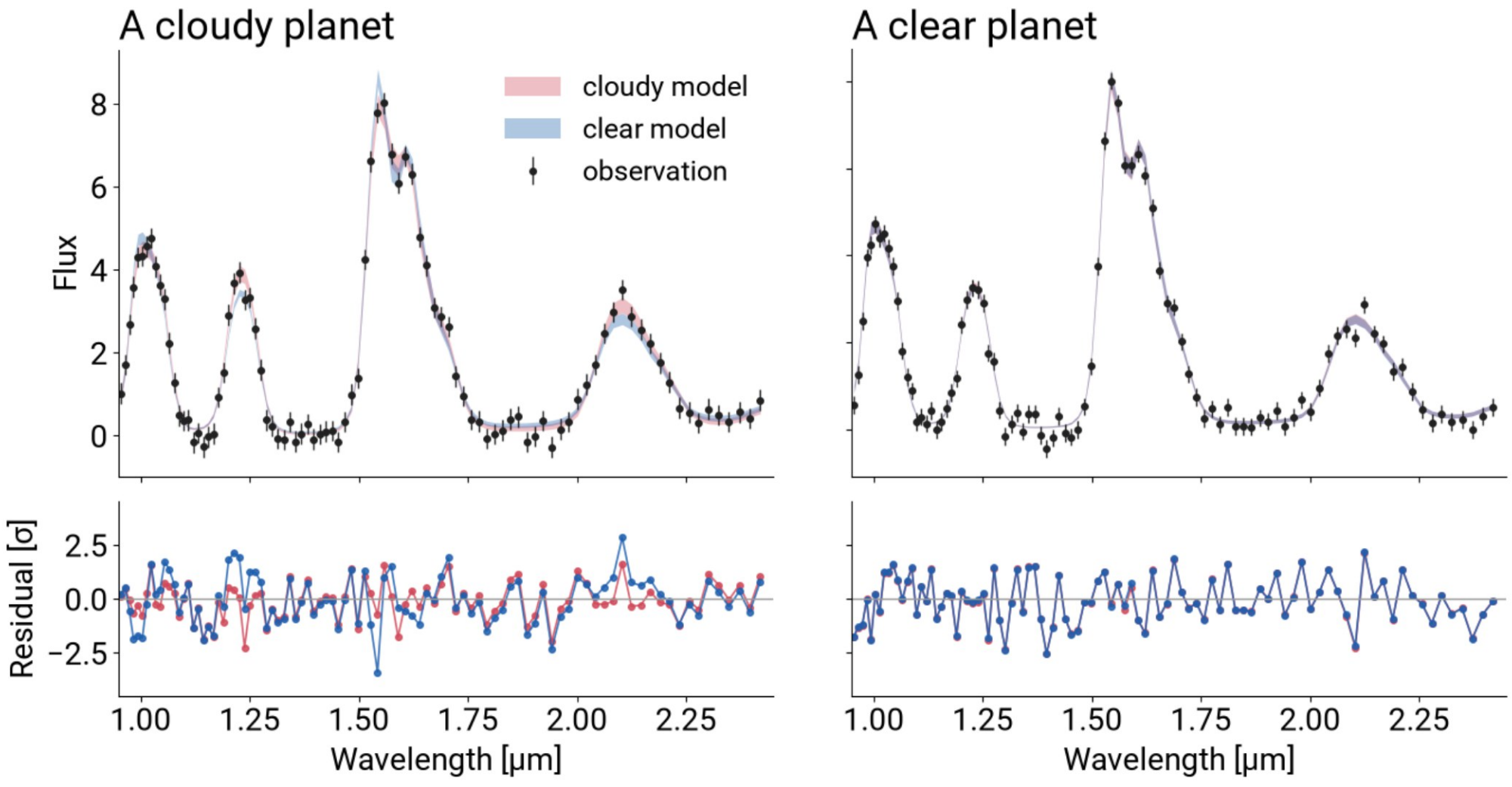
Weighted samples follow the **exact** posterior. The efficiency $\varepsilon = (\sum_i w_i)^2 / (n \sum_i w_i^2)$ flags a poor network. The mean weight estimates the **evidence** $p(x)$, needed for model comparison.

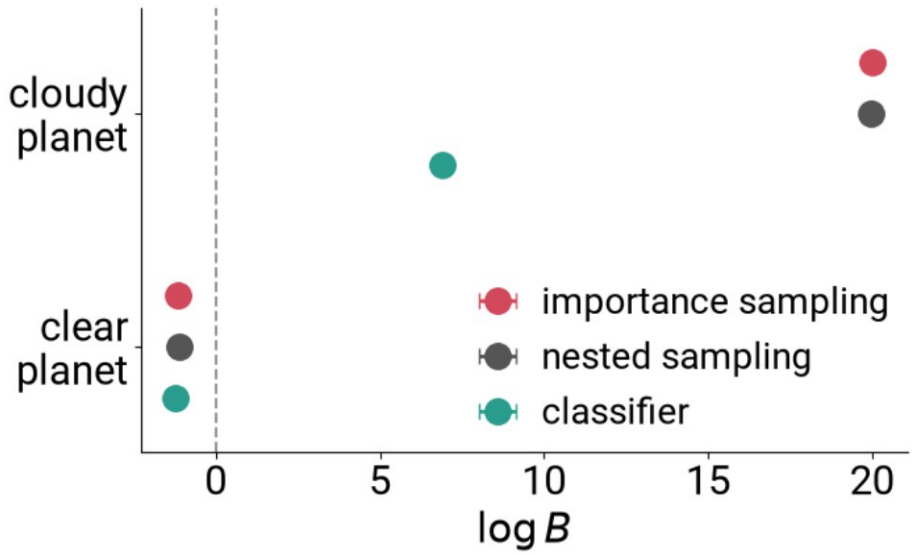


With importance sampling (blue), NPE matches MCMC (grey).

Model comparison

Are there clouds? Compare two simulators, with and without a cloud deck. By eye, both fit.





The **Bayes factor** $B = p(x|\text{cloudy}) \,/\, p(x|\text{clear})$ compares the evidence of the two models. Three ways to estimate it:

Nested sampling $$p(x|M) \approx \sum_i p(x|\theta_i, M)\, \Delta X_i$$

Importance sampling $$p(x|M) \approx \frac{1}{n} \sum_i \frac{p(x|\theta_i, M)\, p(\theta_i|M)}{q_\phi(\theta_i|x)}, \quad \theta_i \sim q_\phi(\theta|x)$$

Classifier $$\log B \approx \log \frac{d_\psi(x)}{1 - d_\psi(x)}, \quad d_\psi(x) \approx p(\text{cloudy}|x)$$

Classifier: Radev et al, 2020. Model comparison without the evidence: MIRA, Sharief et al, ICML 2026.

Summary

Inference requires a simulator, not a likelihood.

A neural posterior estimator is trained once; it then gives the posterior of any spectrum in seconds.

Calibration tests the network; posterior predictive checks test the model.

An overdispersed posterior is safer than an overconfident one.

With a likelihood, importance sampling corrects the network and gives the evidence, for model comparison.

Reading list

Reviews

- Dax et al, An introduction to Bayesian and frequentist simulation-based inference with machine learning, 2026.
- Cranmer, Brehmer and Louppe, The frontier of simulation-based inference, PNAS 2020.
- Deistler et al, Simulation-based inference: a practical guide, 2025.
- Arruda et al, Diffusion models in simulation-based inference: a tutorial review, TMLR 2026.

Simulation-based inference for exoplanets

- Márquez-Neila et al, Supervised machine learning for analysing spectra [...], Nature Astronomy 2018.
- Haldemann et al, Exoplanet characterization using conditional invertible neural networks, A&A 2023.
- Vasist et al, Neural posterior estimation for exoplanetary atmospheric retrieval, A&A 2023.
- Aubin et al, Simulation-based inference for exoplanet atmospheric retrieval [...], 2023.
- Ardévol Martínez et al, FlopPITy: enabling self-consistent exoplanet atmospheric retrievals [...], A&A 2024.
- Gebhard et al, Flow matching for atmospheric retrieval of exoplanets, A&A 2025.
- Lueber et al, Near-instantaneous atmospheric retrievals and model comparison with FASTER, ApJL 2025.
- Vasist et al, Panchromatic characterization of the Y0 brown dwarf WISE 1738 using JWST/MIRI, A&A 2025.



The end.